

Notes House

Your Study Companion

NOTES HOUSE

A SMART ACADEMIC PLATFORM FOR UNIVERSITY STUDENTS IN PAKISTAN

PRESENTED BY

HASNAT AHMAD	076421	2022-GCG-123
ABDUL HANAN	076329	2022-GCG-126
HAFIZ ABDUL REHMAN	076370	2022-GCG-149

SUPERVISOR

SIR KASHIF RIAZ

INTRODUCTION

- **Project is**

A mobile application that provides a centralized platform for students across Pakistan to access notes, past papers, and course outlines in one place. It helps save time, reduces stress, and makes study material easily accessible and organized.

- **Unlike**

Unlike traditional methods where students search for study material in WhatsApp groups, old drives, or social media, Notes House offers a structured, reliable, and verified system where all content is properly managed and easy to find.

- **At the core**

At the core, the system is based on a multi-role structure where teachers upload verified content, admins approve it, and students access it. It also includes features like quizzes, in-app PDF reading, protected downloads, and an AI chatbot for better learning support.

OBJECTIVES OF THE PROPOSED SYSTEM

- **Multi-role System**

Provide a system with Student, Teacher, and Admin roles for controlled and verified content sharing.

- **Retrieval Efficiency**

Enable fast and easy access to notes, past papers, and outlines from a centralized platform.

- **Knowledge Enhancement**

Support learning through quizzes, performance analysis, and AI-based question answering.

- **User-Friendly Experience**

Design a simple, intuitive interface with built-in PDF reader and smooth navigation.

- **Scalable & Secure System**

Ensure secure data storage, protected downloads, and support for multiple universities.

- **What (Overall Goal)**

Reduce study stress by providing reliable, organized, and verified study resources in one place.

SCOPE

Notes House covers a centralized platform for students across different universities in Pakistan to access organized notes, past papers, and course outlines. The system allows verified teachers to upload study material, which is reviewed and approved by admins to ensure quality and authenticity.
It includes a built-in PDF reader for viewing documents within the app and supports secure in-app downloads to protect content from external sharing. The platform also provides quizzes and basic performance analysis to help students evaluate their preparation.
Additionally, an AI-based chat feature assists users in understanding topics directly from the study material. The system ensures secure data storage, proper access control, and a simple, user-friendly interface for smooth navigation.

RELATED SYSTEM ANALYSIS / LITERATURE REVIEW

- **Challenges (Existing Apps)**

Existing platforms like Past Papers Pakistan and Ilmkidunya provide limited and often unverified content. Many resources are outdated, incomplete, and not properly organized according to semesters or universities.

- **Current Systems**

Most current apps lack features like verified uploads, admin control, quizzes, and structured study material. Some also restrict access or require subscriptions, reducing usability for students.

- **Proposed System Perspective**

Notes House overcomes these limitations by providing verified notes, past papers, and outlines in one place. It includes admin approval, quiz system, AI support, built-in PDF reader, and secure in-app downloads for a complete learning solution.

TOOLS AND TECHNOLOGIES

- **Development Tools**

Flutter SDK (3.x), Android Studio, VS Code

- **Programming Language**

Dart (Latest Stable)

- **Backend & Database**

Node.js with Express.js (Local Server), Firebase (Authentication & optional services)

- **Storage Services**

Local-Host Storage (for notes, past papers, PDFs)

- **APIs & Integration**

REST APIs using Node.js & Express.js (for data handling), AI/LLM API (for chatbot feature)

- **Additional Libraries**

PDF Viewer (Flutter PDF Viewer / Syncfusion)

FUNCTIONAL REQUIREMENT

- **User Authentication**

User registration, login, password recovery, and secure session management for students and teachers.

- **Study Material Access**

View organized notes, past papers, and outlines with search, filter, in-app PDF viewing, and secure downloads.

- **Teacher Upload System**

Teachers can upload study material which is reviewed and approved by admin before publishing.

- **Admin Control Panel**

Admins can approve/reject content, manage users, and monitor system activities.

FUNCTIONAL REQUIREMENT

- **Quiz System**

Students can attempt quizzes, get instant results, and view correct answers.

- **Performance Analysis**

System stores quiz history and shows performance using graphs and reports.

- **AI Chatbot Support**

Users can ask questions from notes or general topics and receive instant AI-based responses.

- **Additional Features**

Includes in-app PDF reader, download manager, and profile/settings management.

NON-FUNCTIONAL REQUIREMENTS

- **Reliability**

System ensures stable performance with minimal crashes, automatic recovery after failure, and proper error handling.

- **Usability**

User-friendly interface with simple navigation, readable design, and easy learning for all types of users.

- **Performance**

Fast response time for loading notes, quizzes, and AI responses with efficient data processing and scalability support.

- **Security**

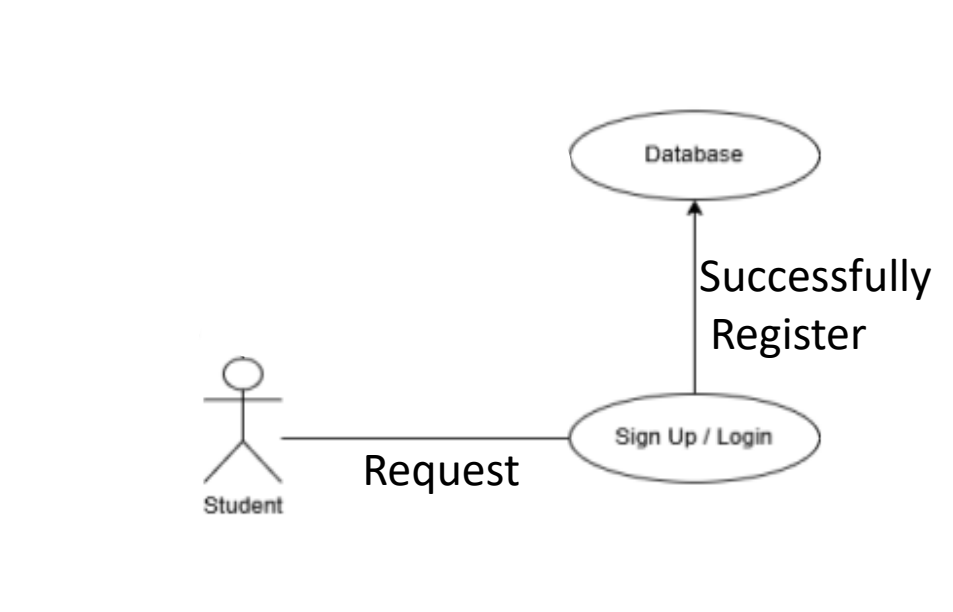
Secure data storage with encryption, role-based access control, and protection against unauthorized access.

- **Scalability**

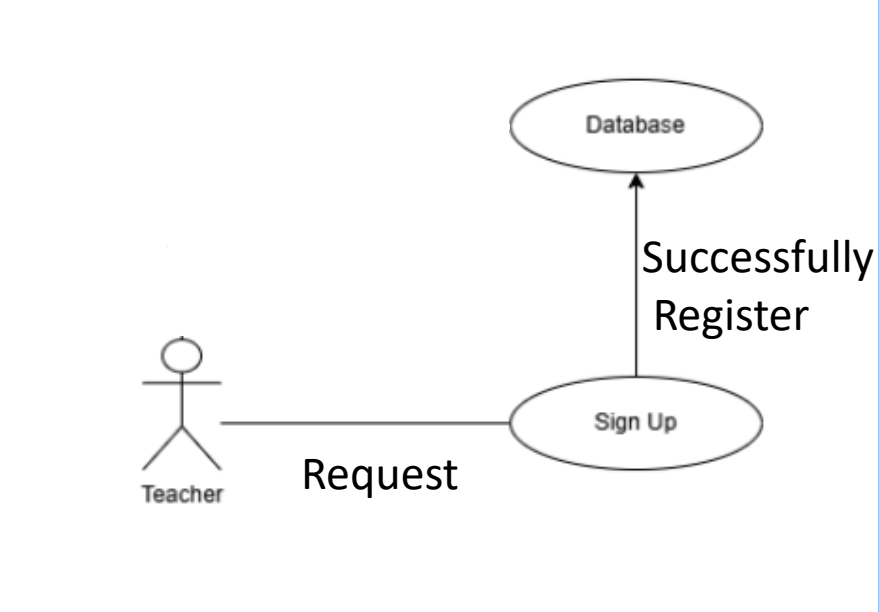
Handles increasing users and data efficiently without affecting performance.

USE CASE DIAGRAMS

- **UC1: Student Sign Up / Login**

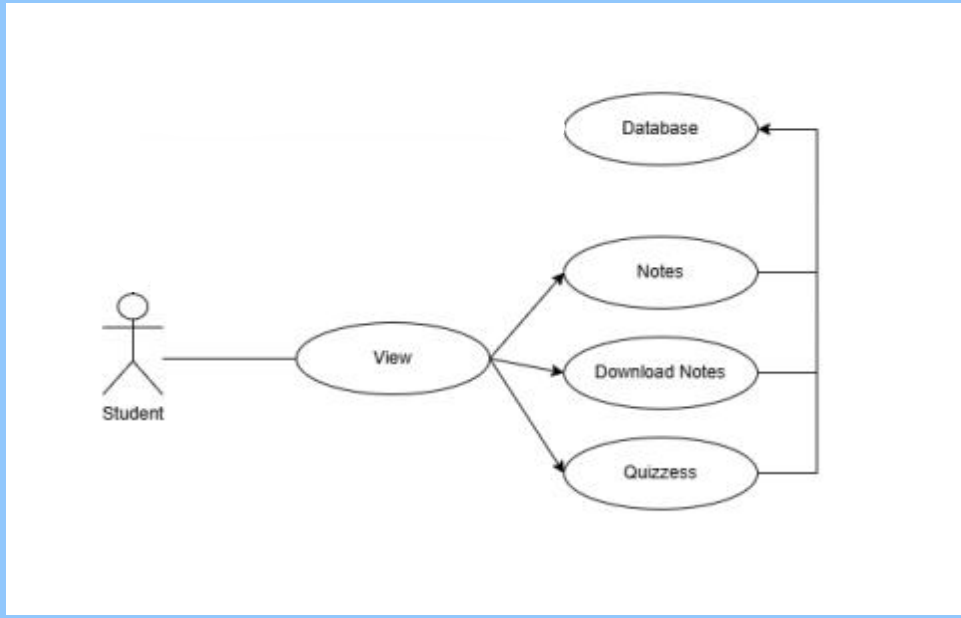


- **UC2: Teacher Sign Up (with edu email verification)**

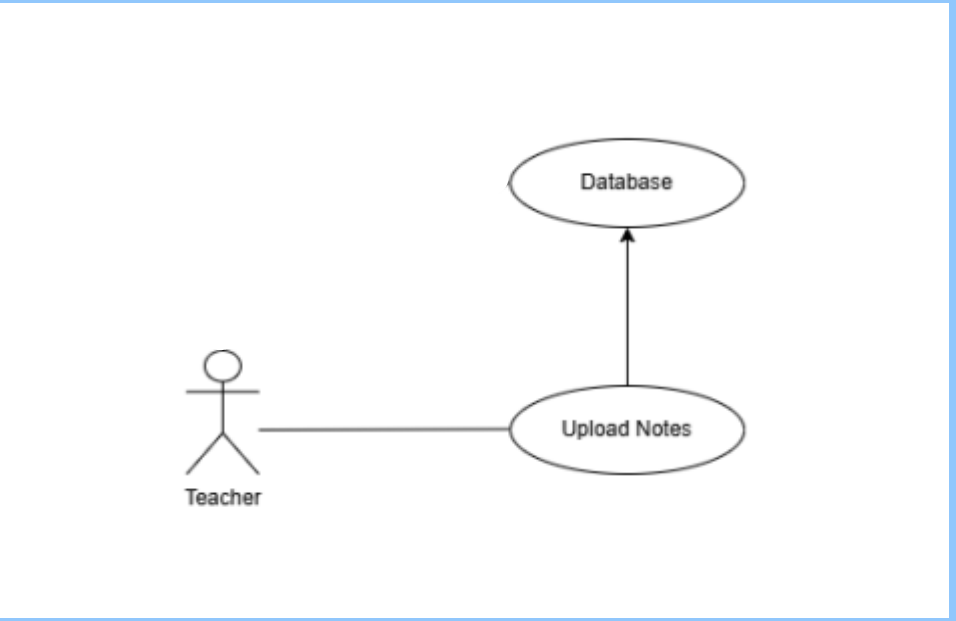


USE CASE DIAGRAMS

- **UC3: View Notes, Downloaded Notes and Quizzes**

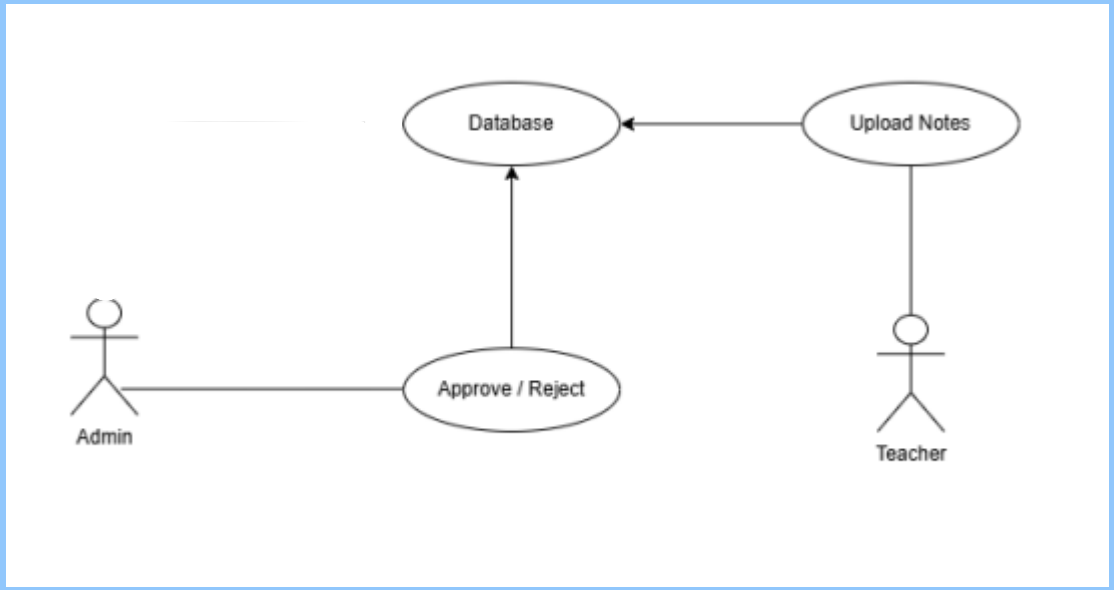


- **UC4: Upload Notes (Teacher)**

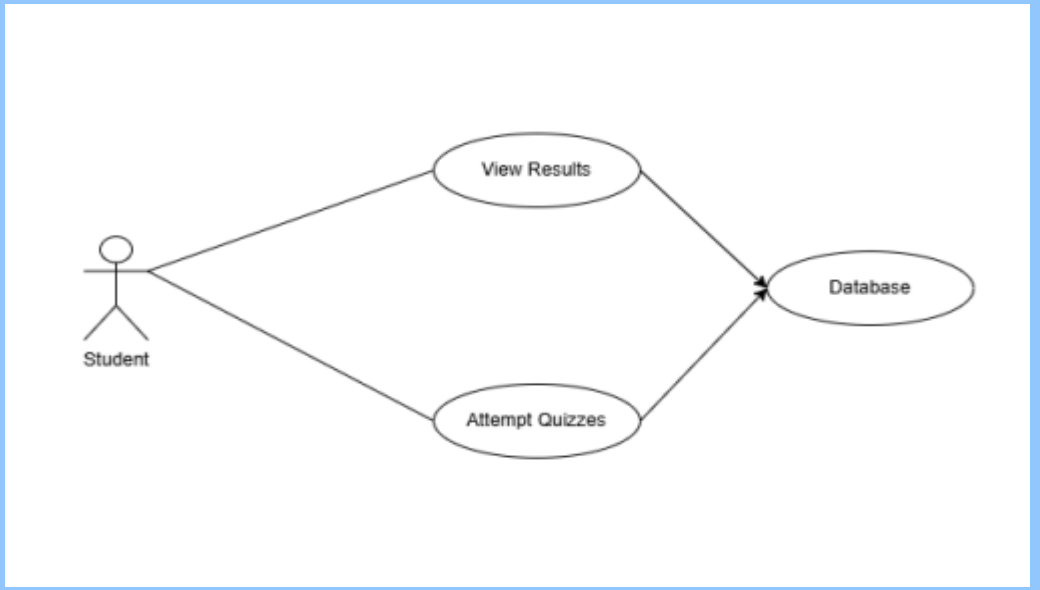


USE CASE DIAGRAMS

- **UC5: Admin Approval / Rejection of Notes**

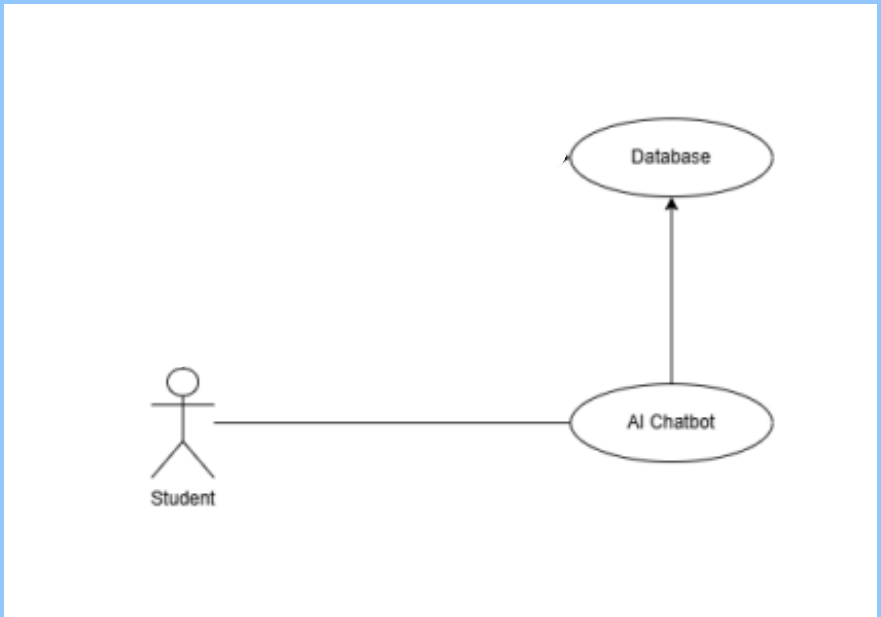


- **UC6: Attempt Quiz and View Results**

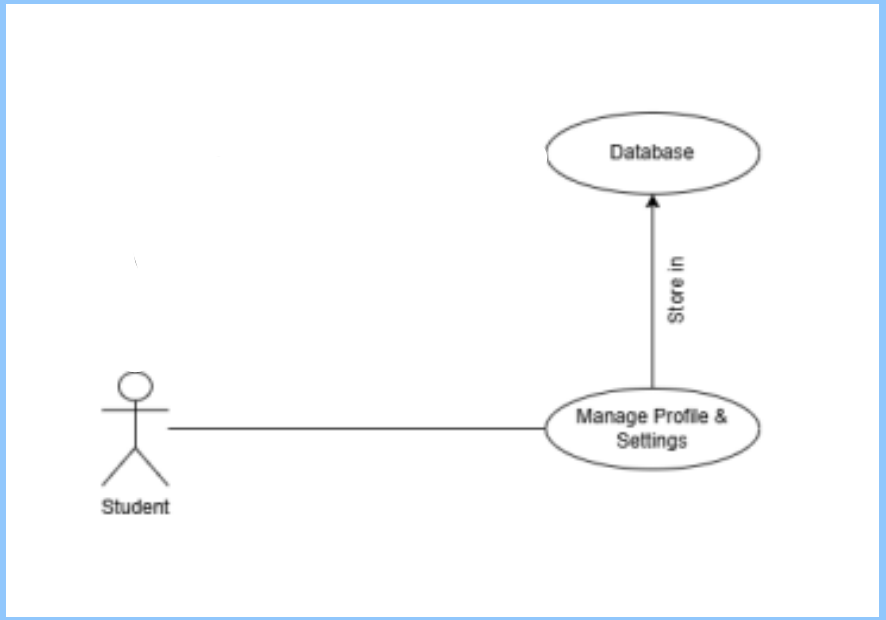


USE CASE DIAGRAMS

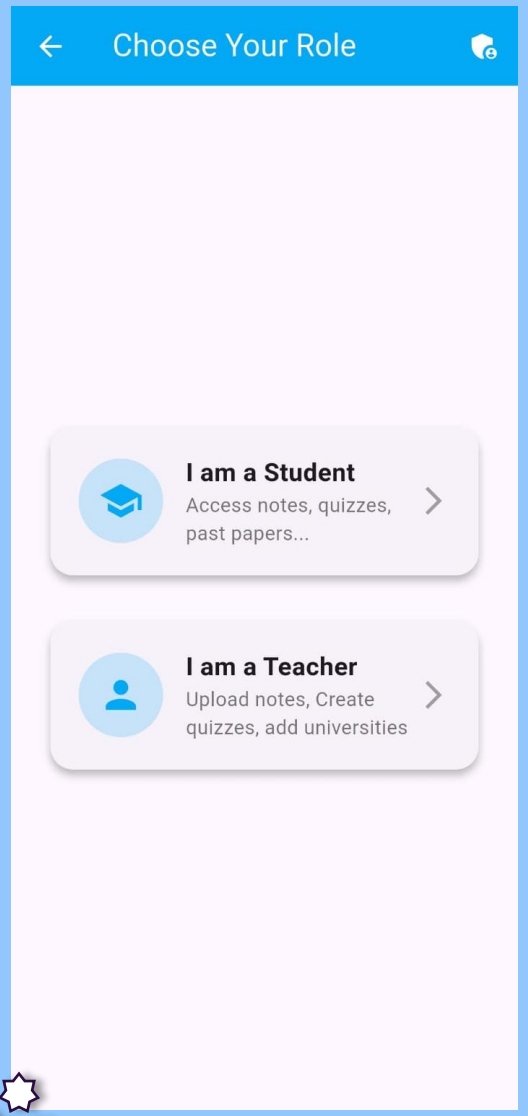
- **UC7: Use AI ChatBot for Study Help**

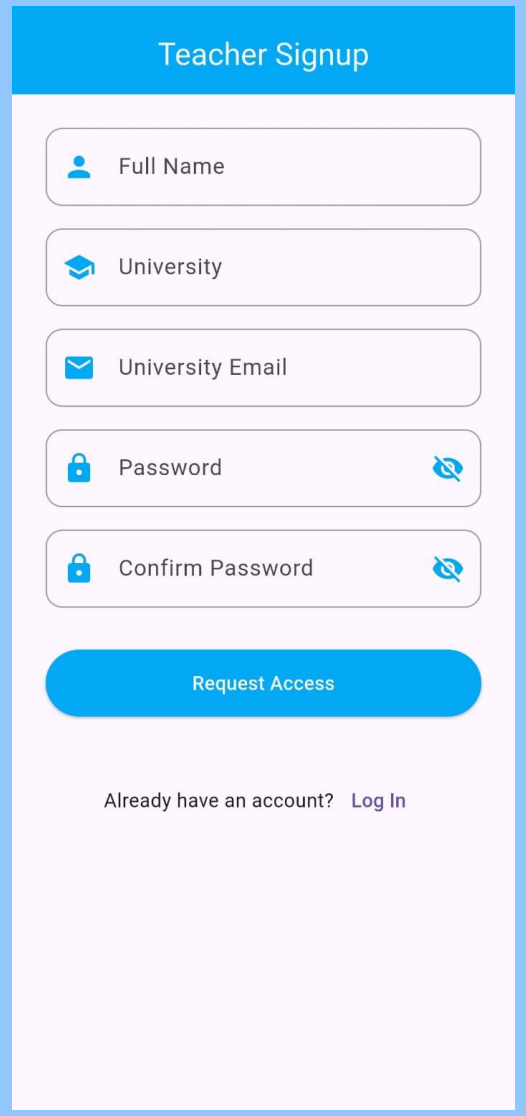


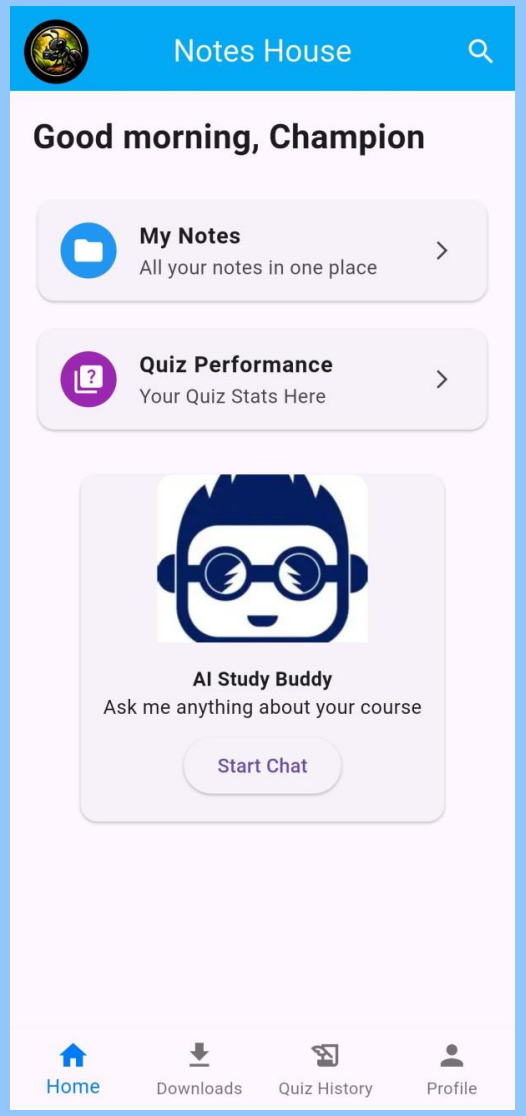
- **UC8: Manage Profile & Settings**

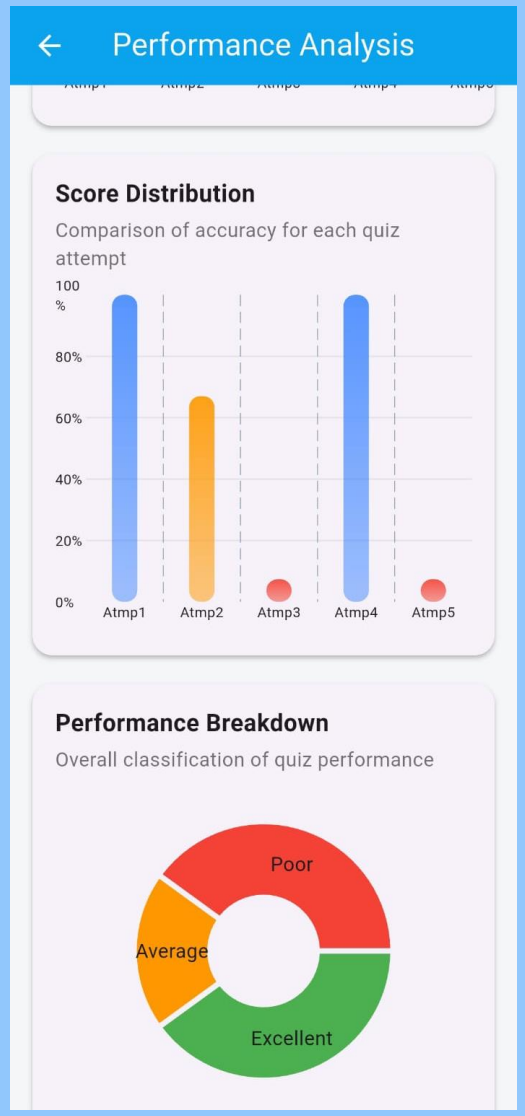


USER SCREENS

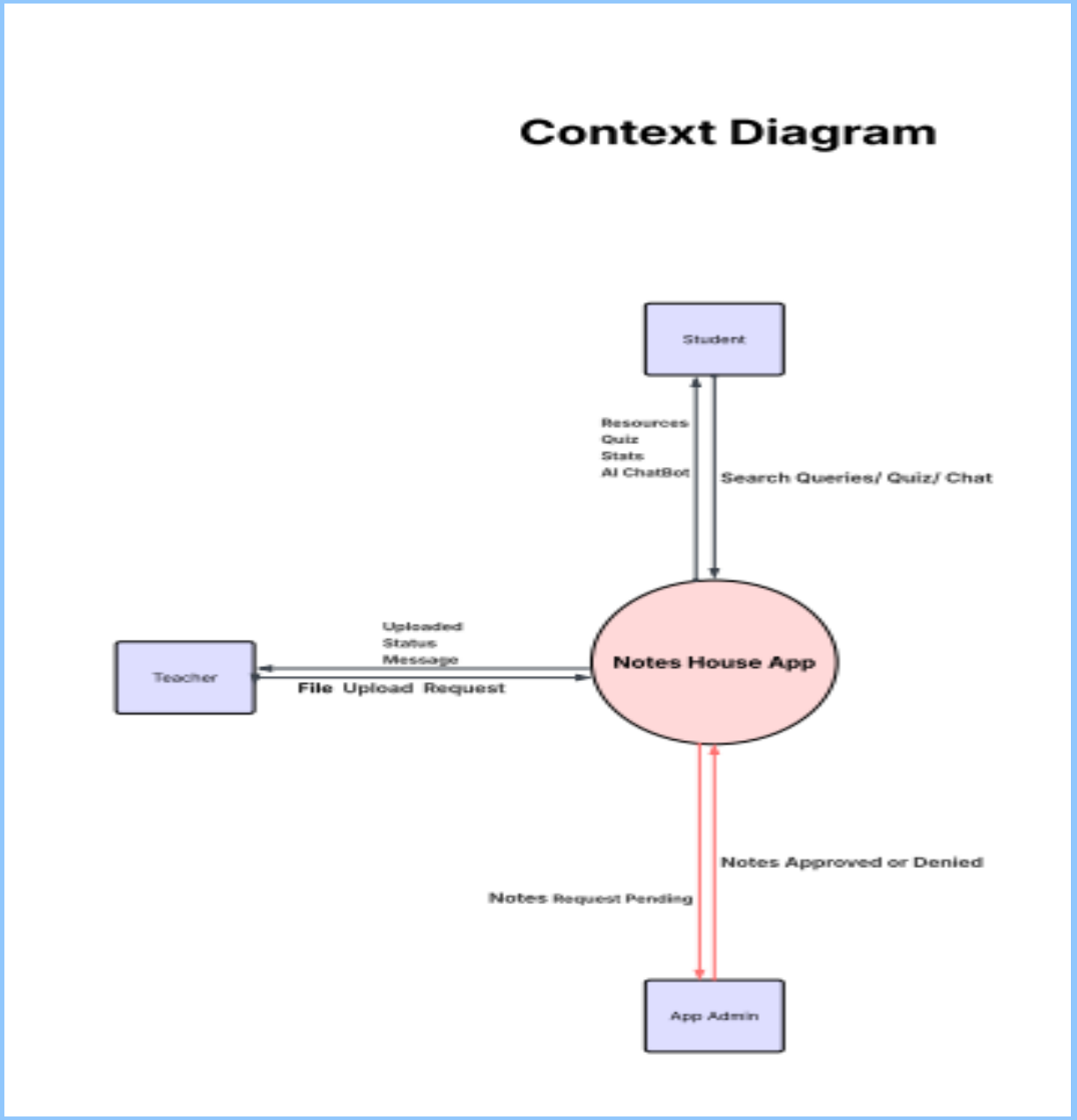




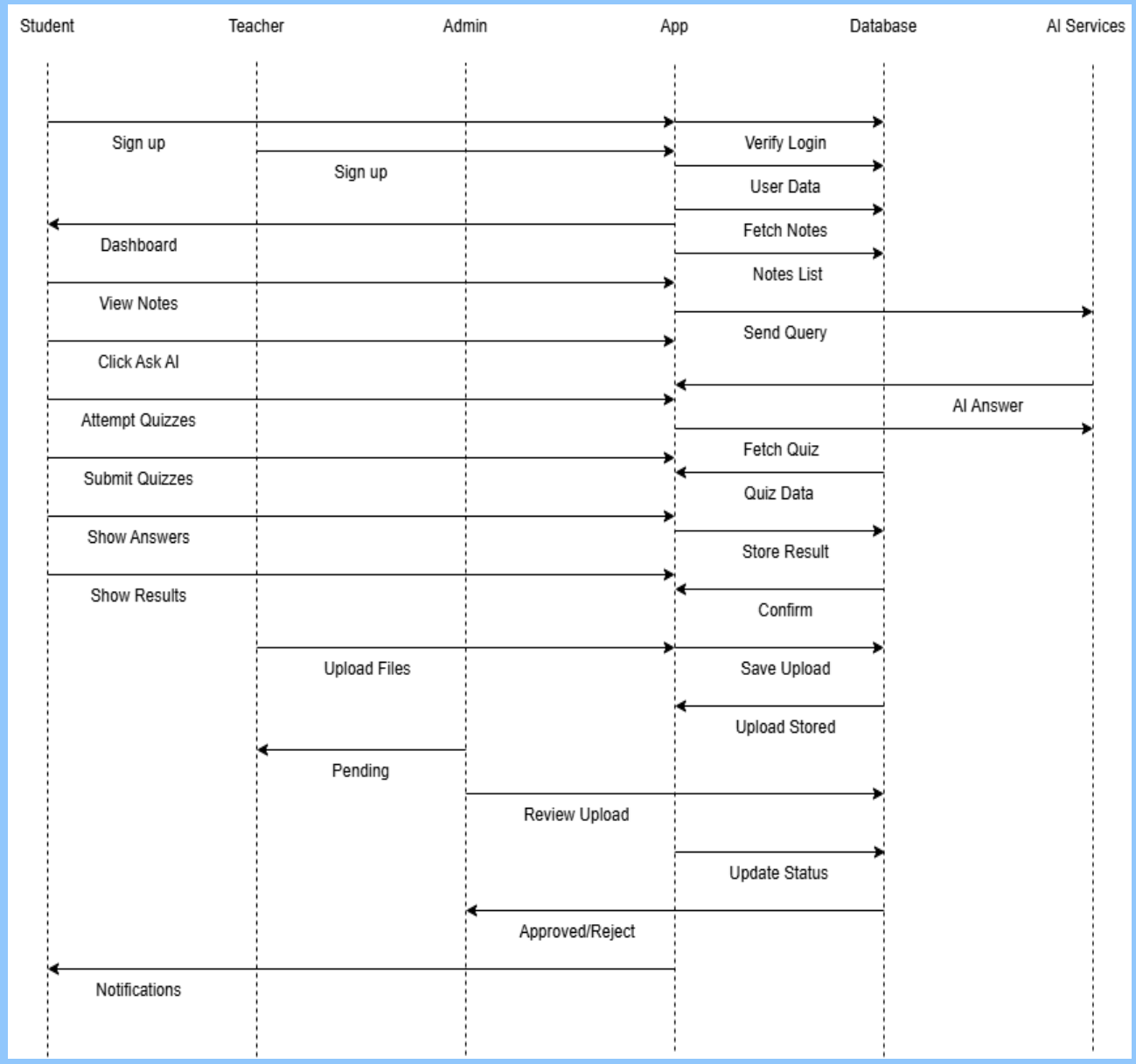




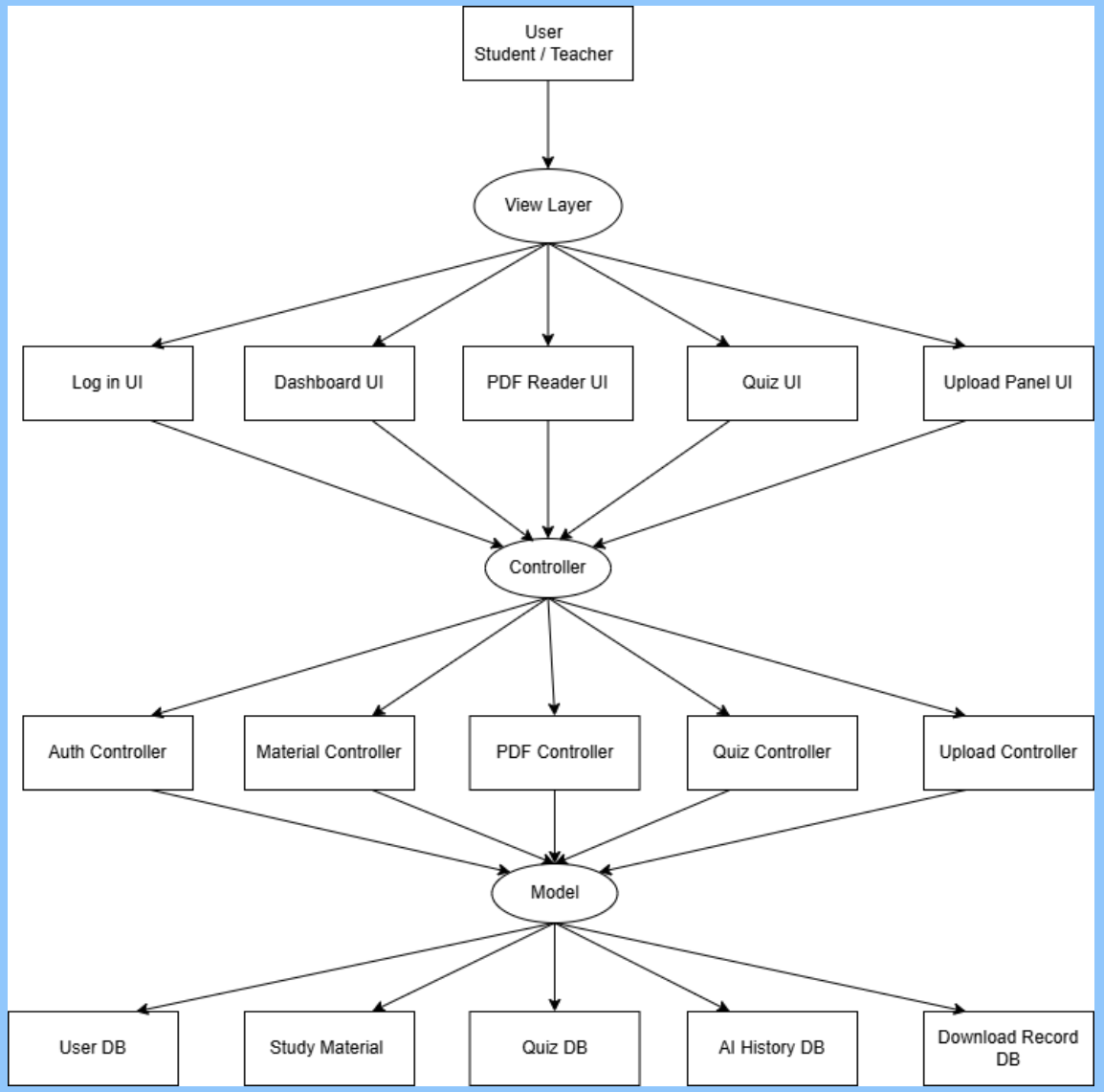
SYSTEM DESIGN



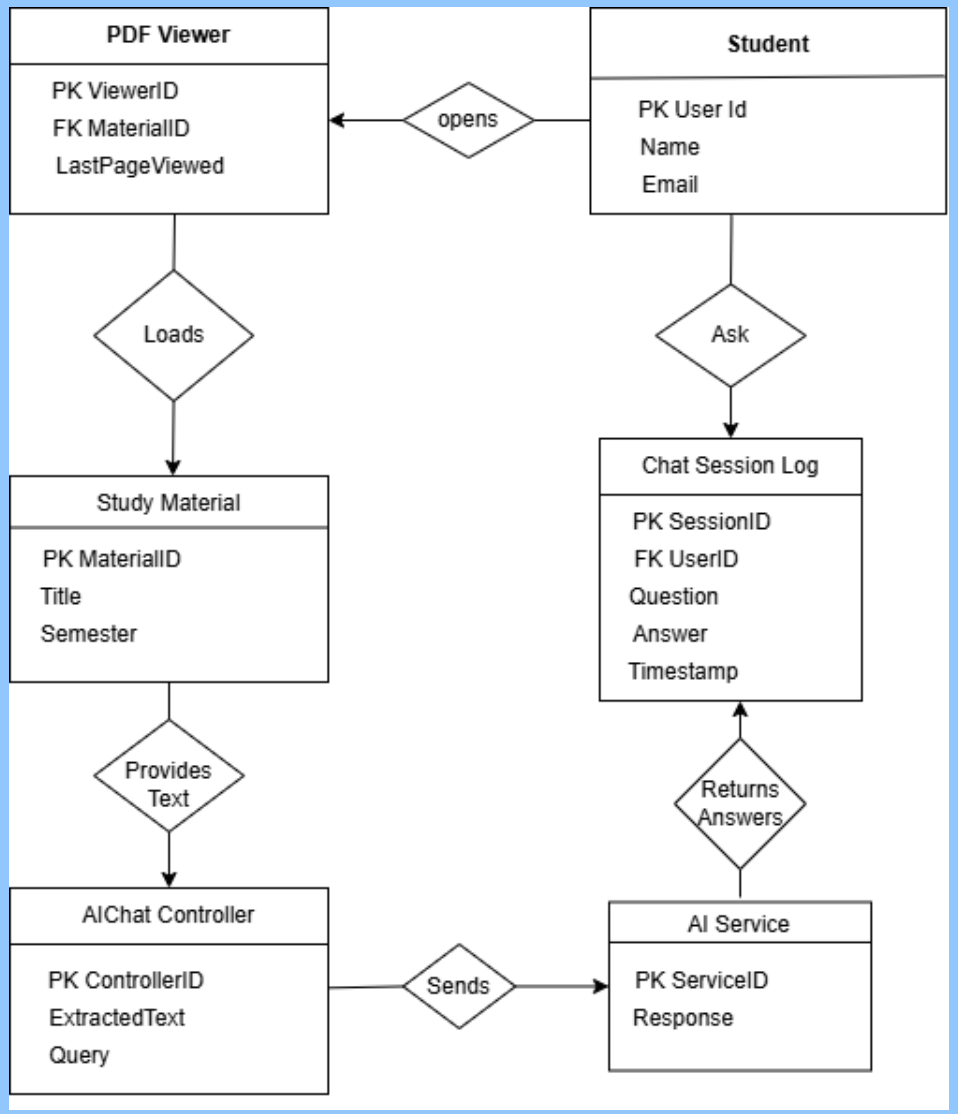
CLASS DIAGRAM



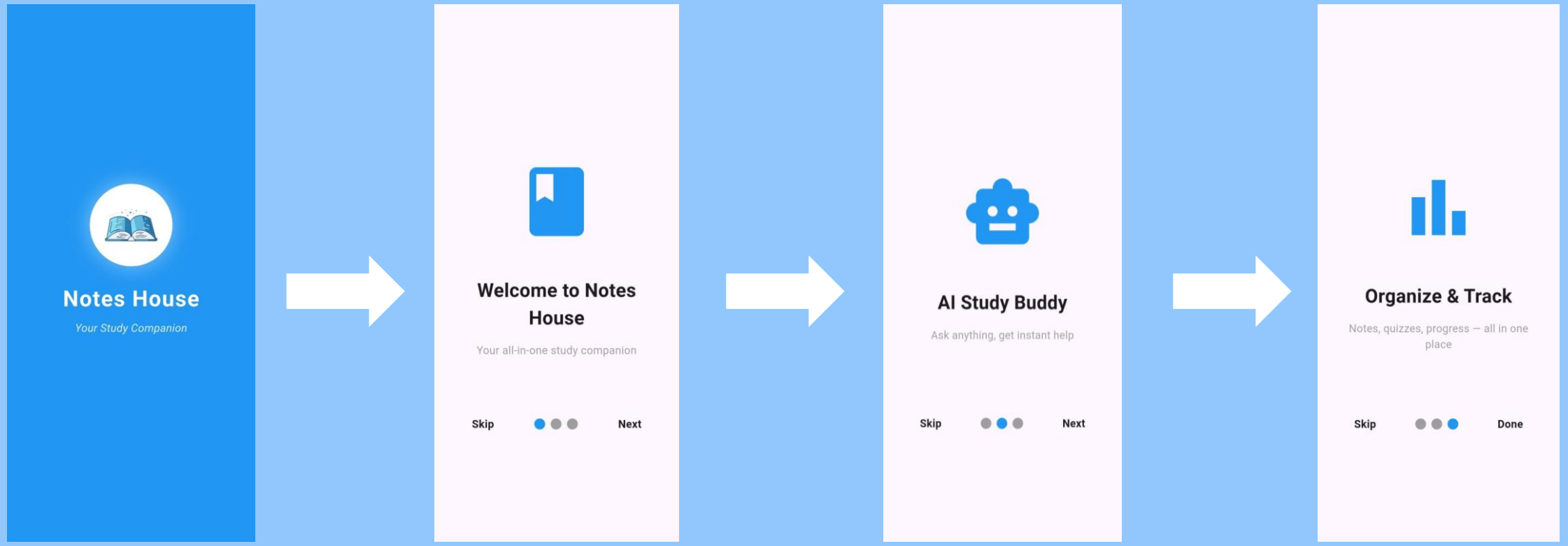
ARCHITECTURAL DIAGRAM



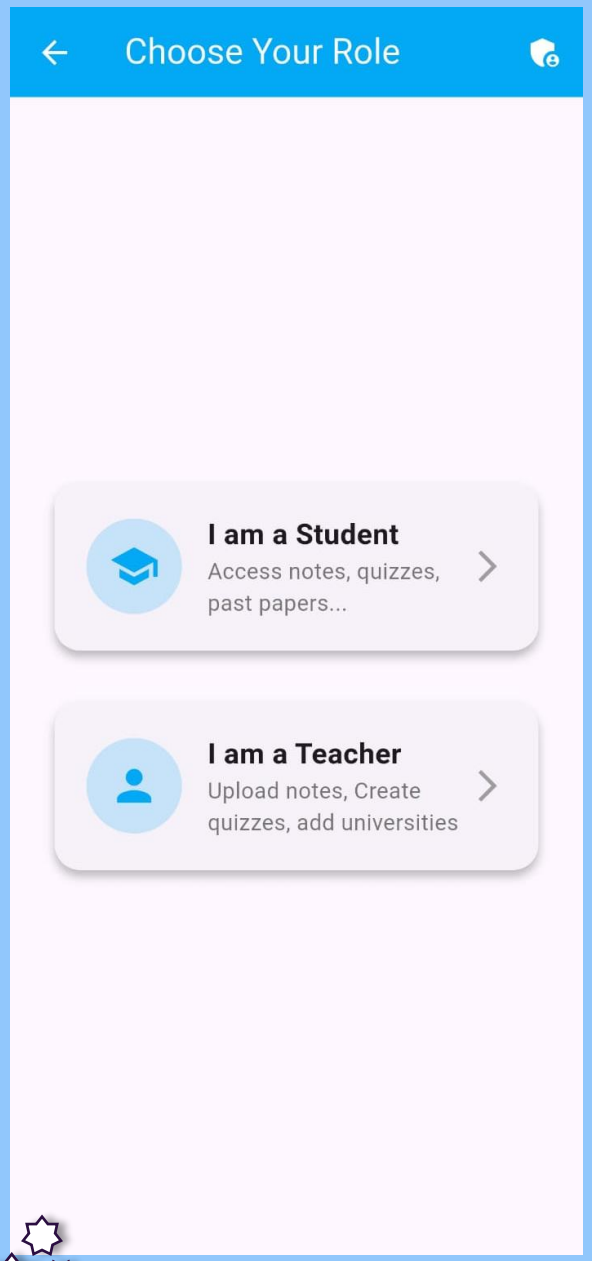
ACTIVITY SEQUENCE DIAGRAM



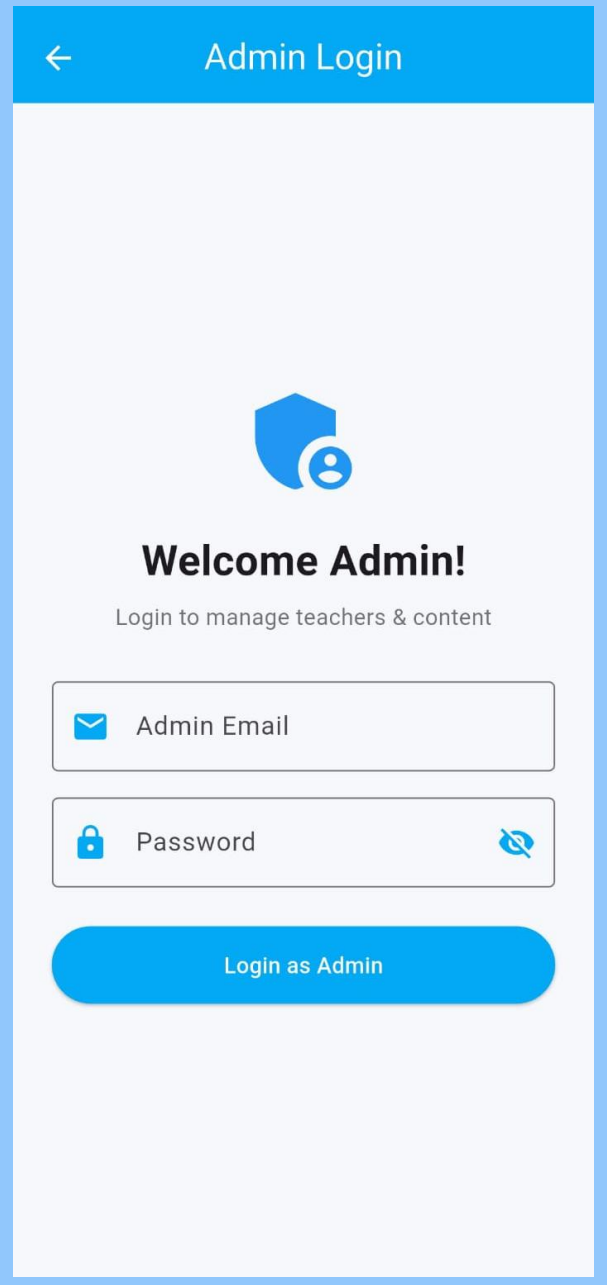
USER INTERFACES



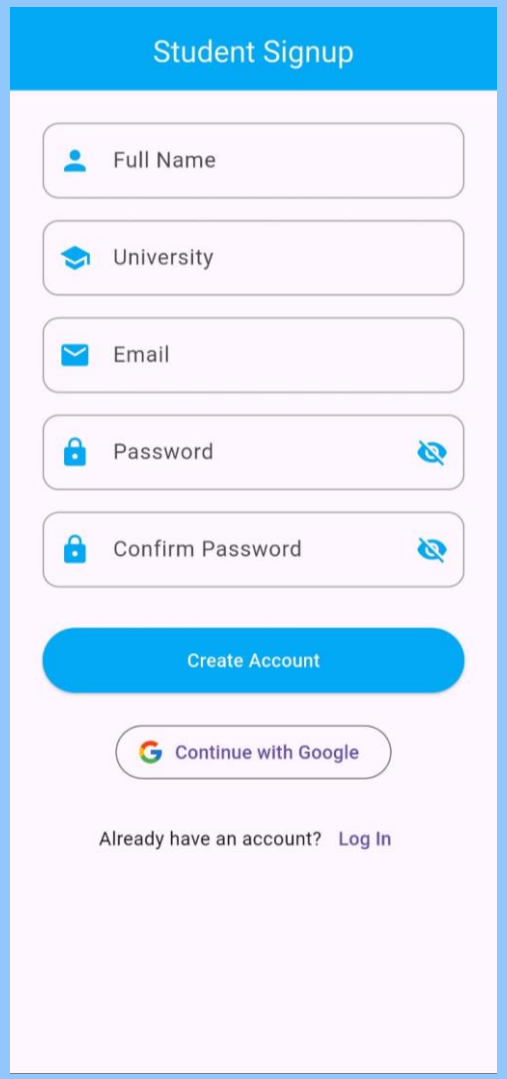
AUTHENTICATION

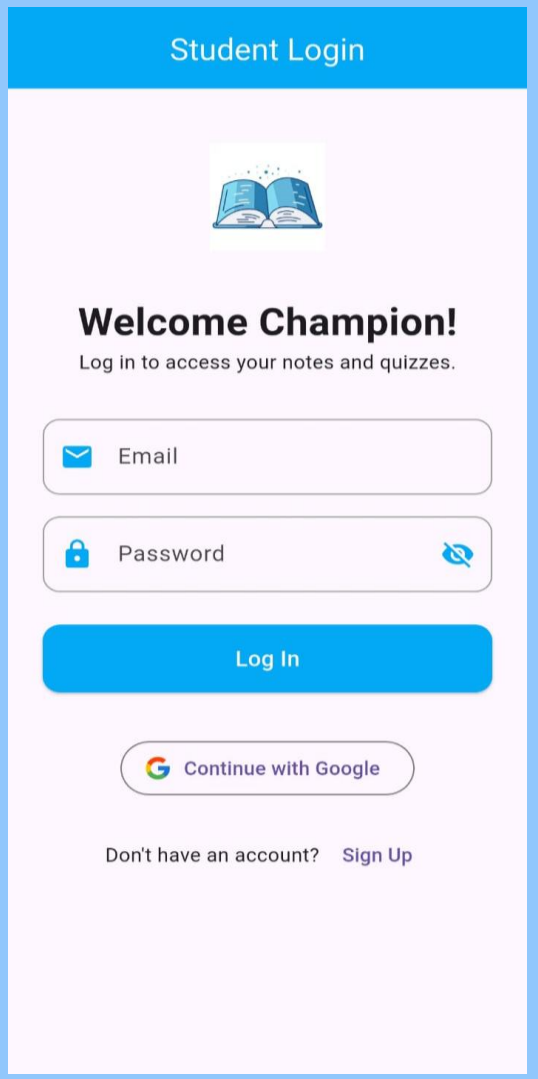


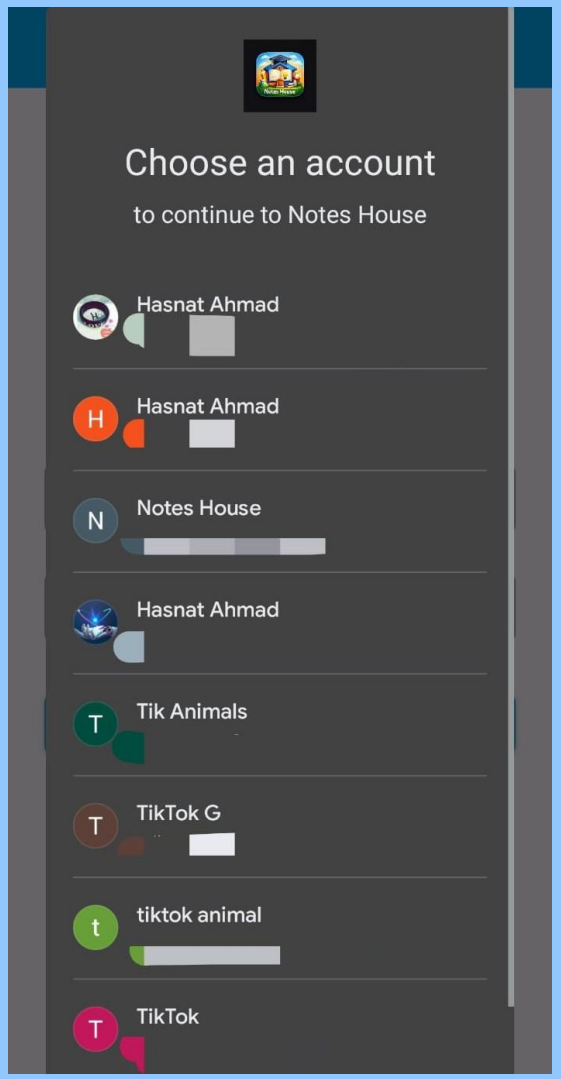
- Secure login and registration system for students and teachers.
- Only verified teachers are allowed to upload content, ensuring reliability and quality of study materials.



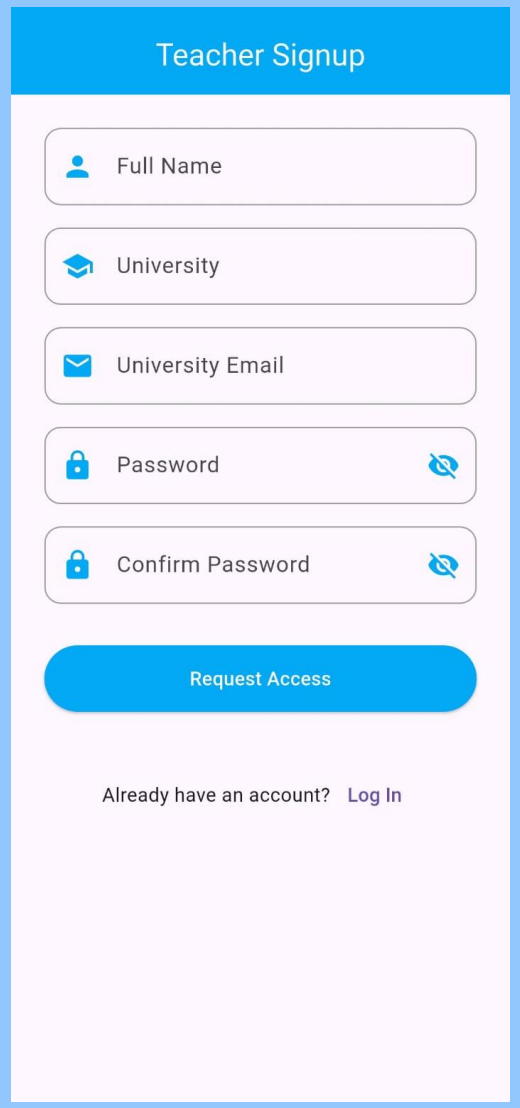
STUDENT AUTHENTICATION

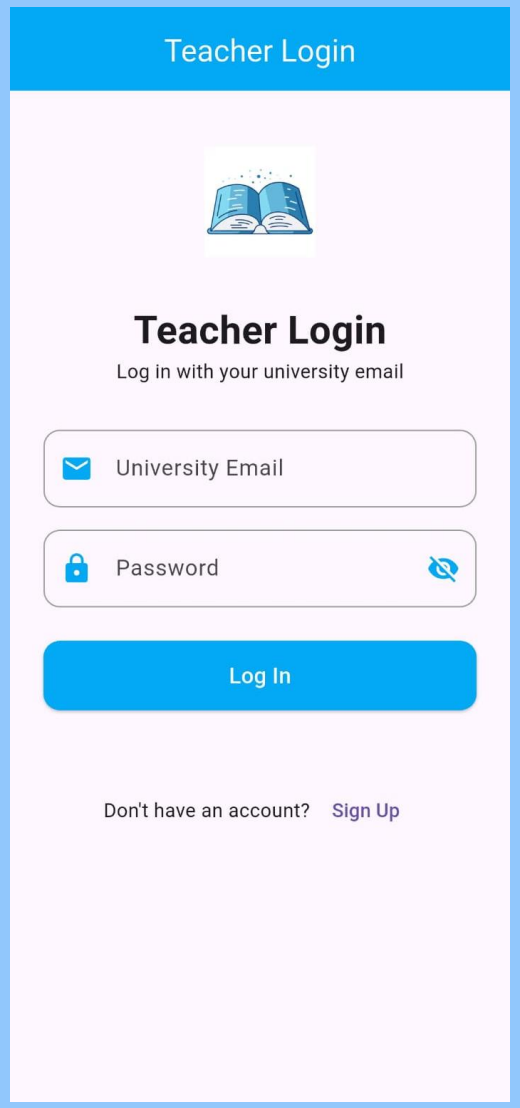


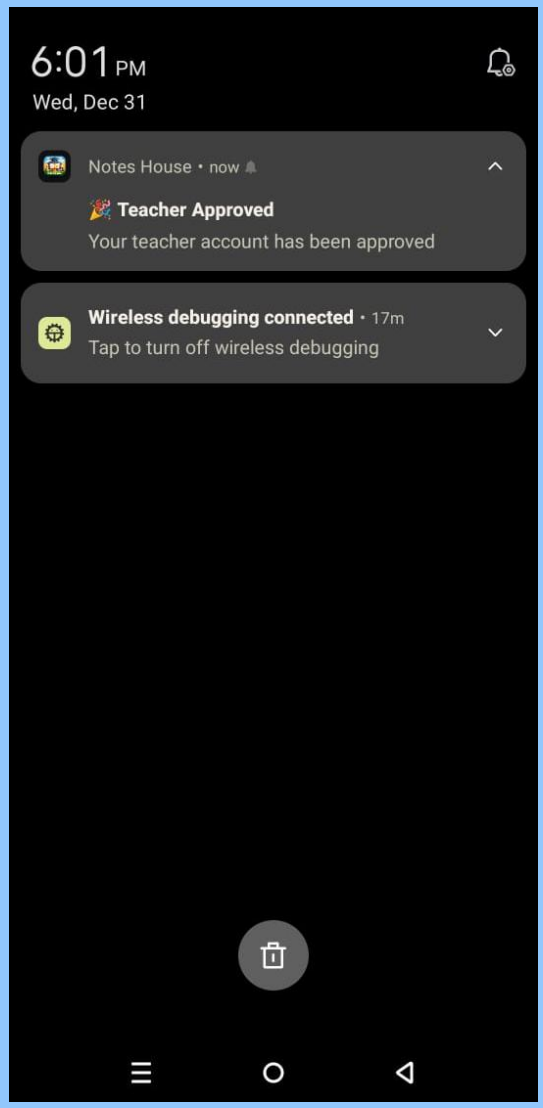




TEACHER AUTHENTICATION



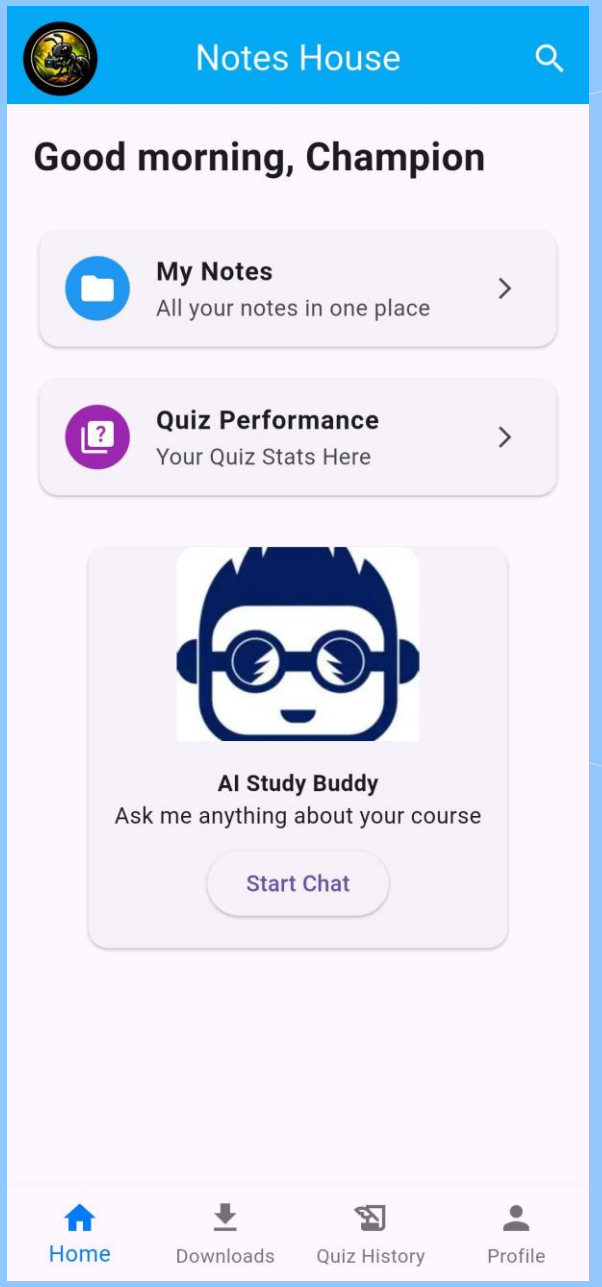




STUDENT DASHBOARD

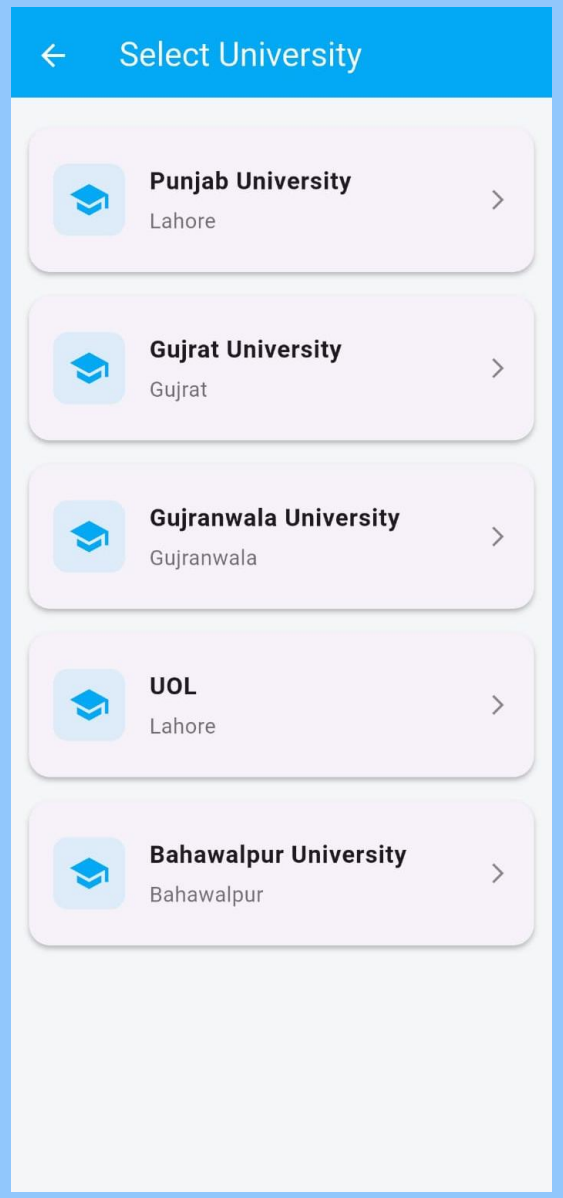
DASHBOARD (HOME-SCREEN)

- Centralized access to all academic resources
- Personalized learning through AI assistance
- Performance tracking via quizzes and analytics.
- Clean and intuitive user interface.

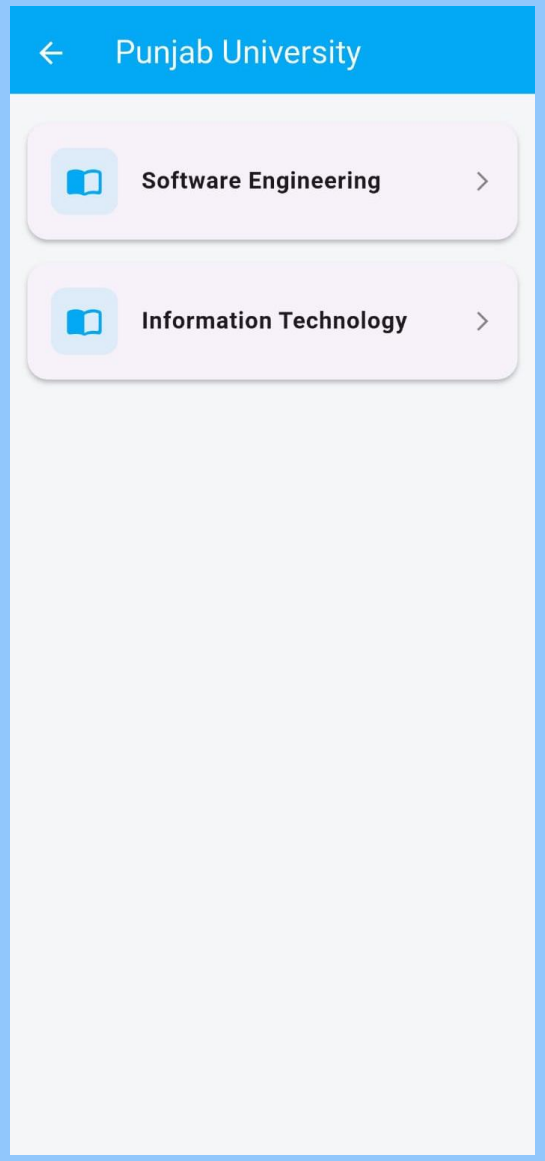


NOTES & STUDY MATERIAL

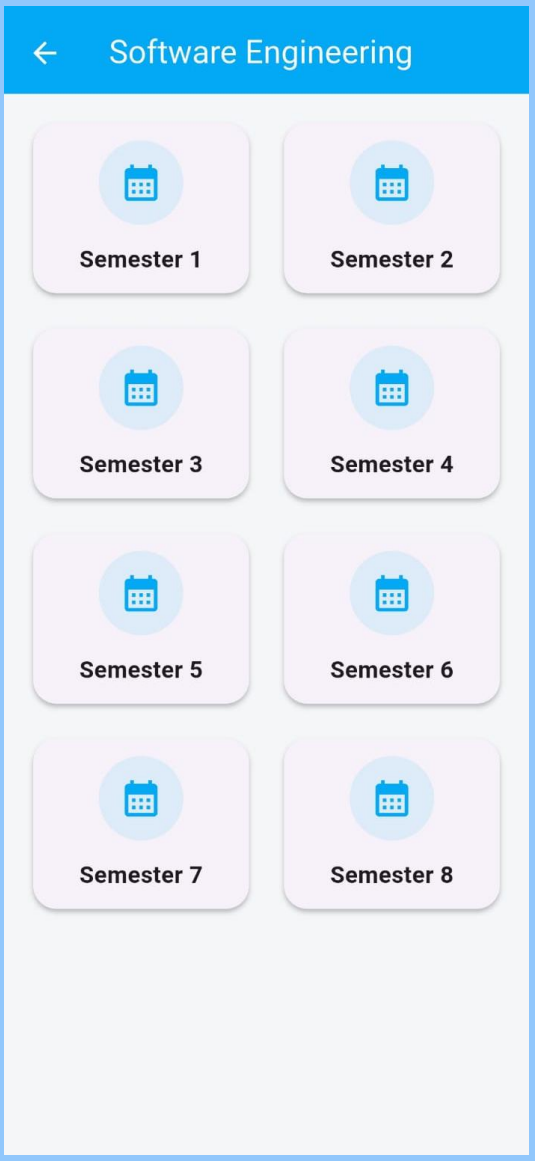
- Provides structured access to notes, books, and past papers organized by:
- **University → Department → Semester → Subject**.
- All content is available in the app without external links.



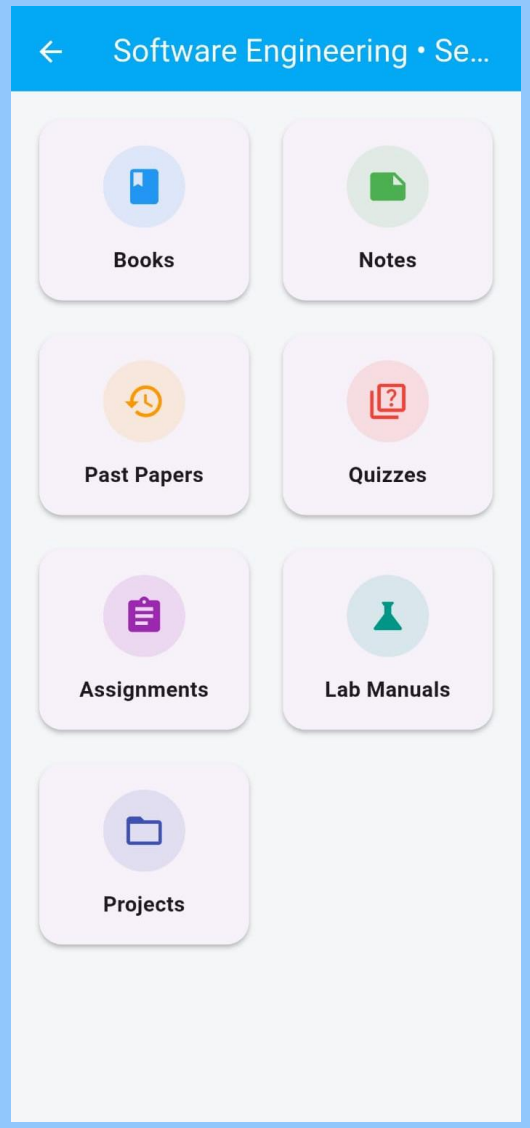
NOTES SELECTION FLOW



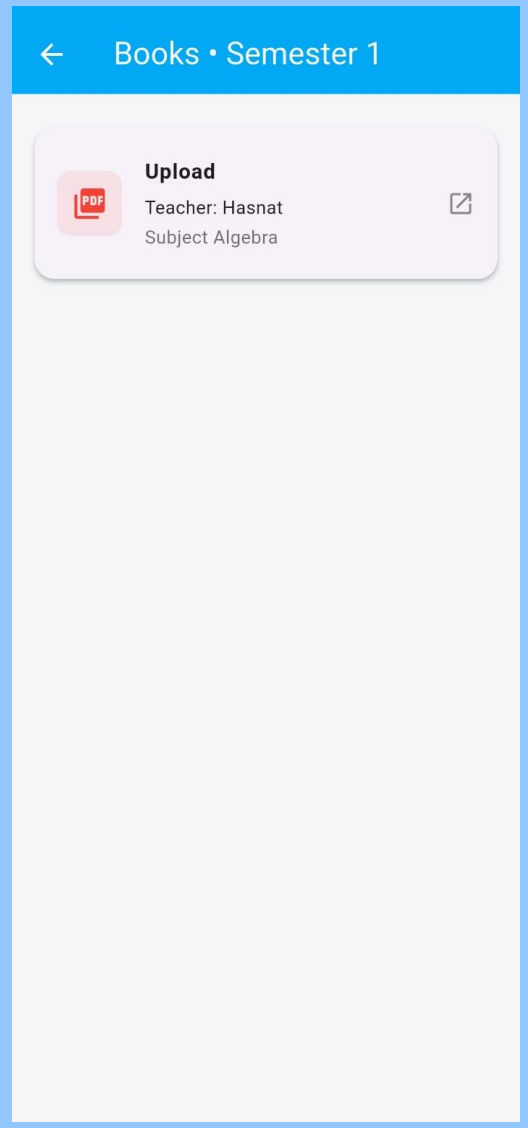




NOTES SELECTION FLOW

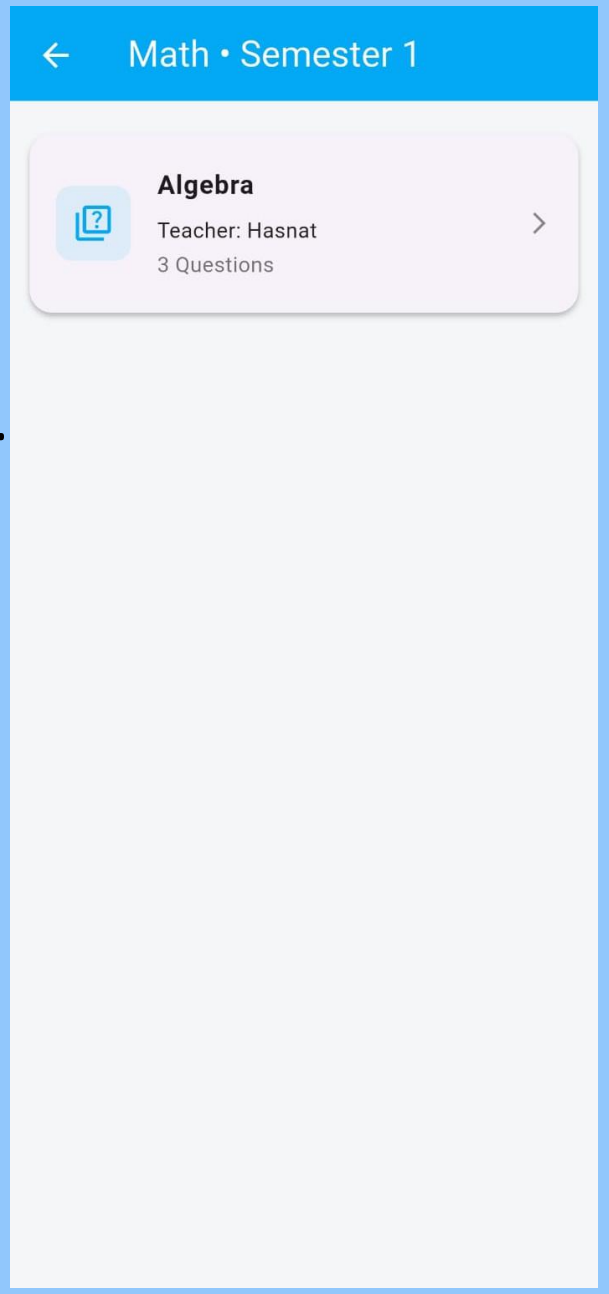




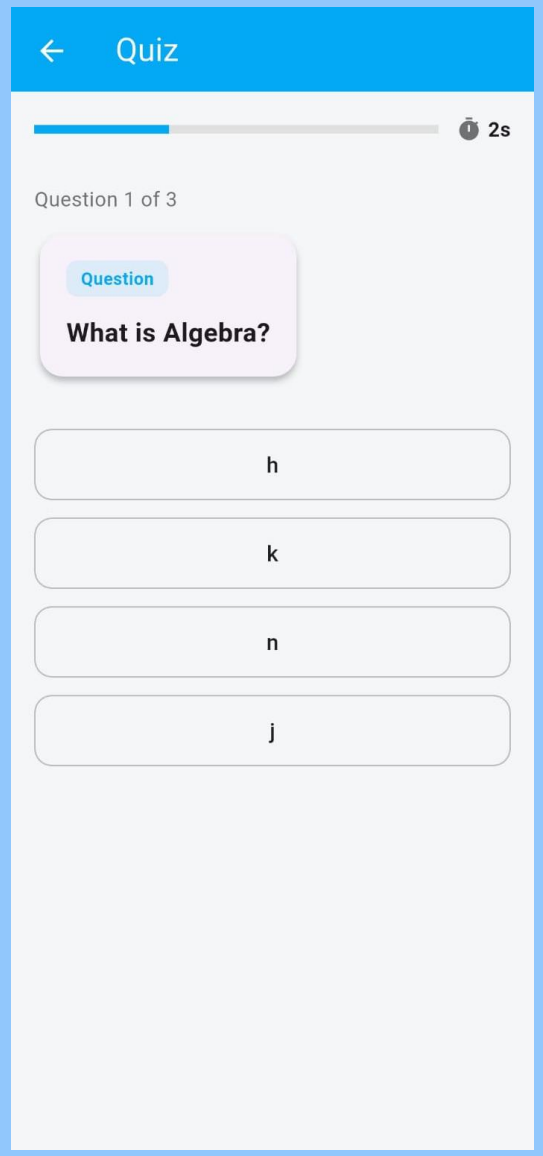


SMART QUIZ SYSTEM

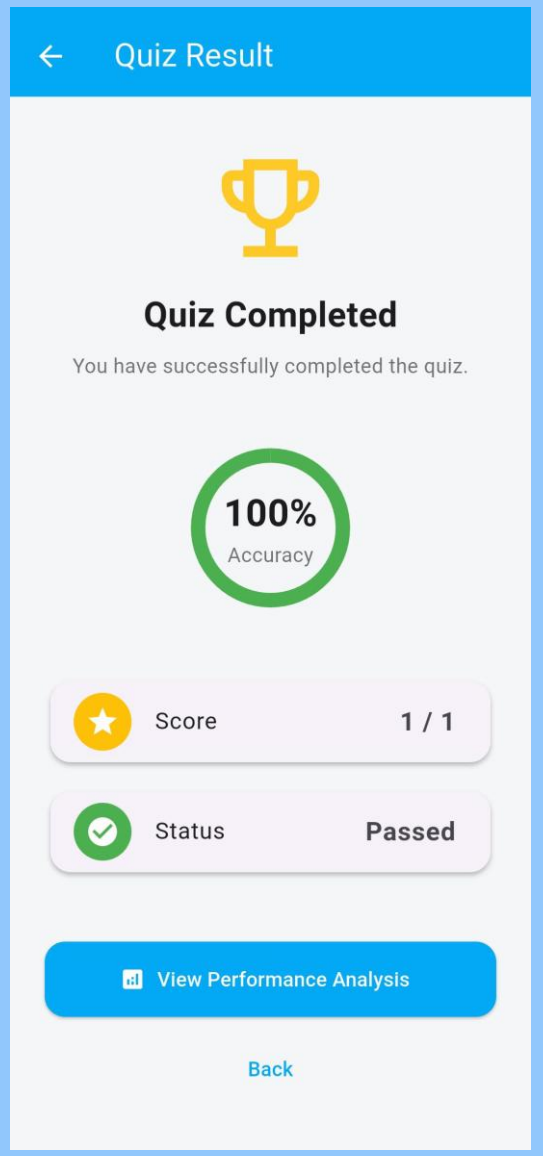
- Teacher-verified quizzes designed to test conceptual understanding.
- Helps students to evaluate their knowledge and improve weak areas.
- Makes learning more interactive and personalized



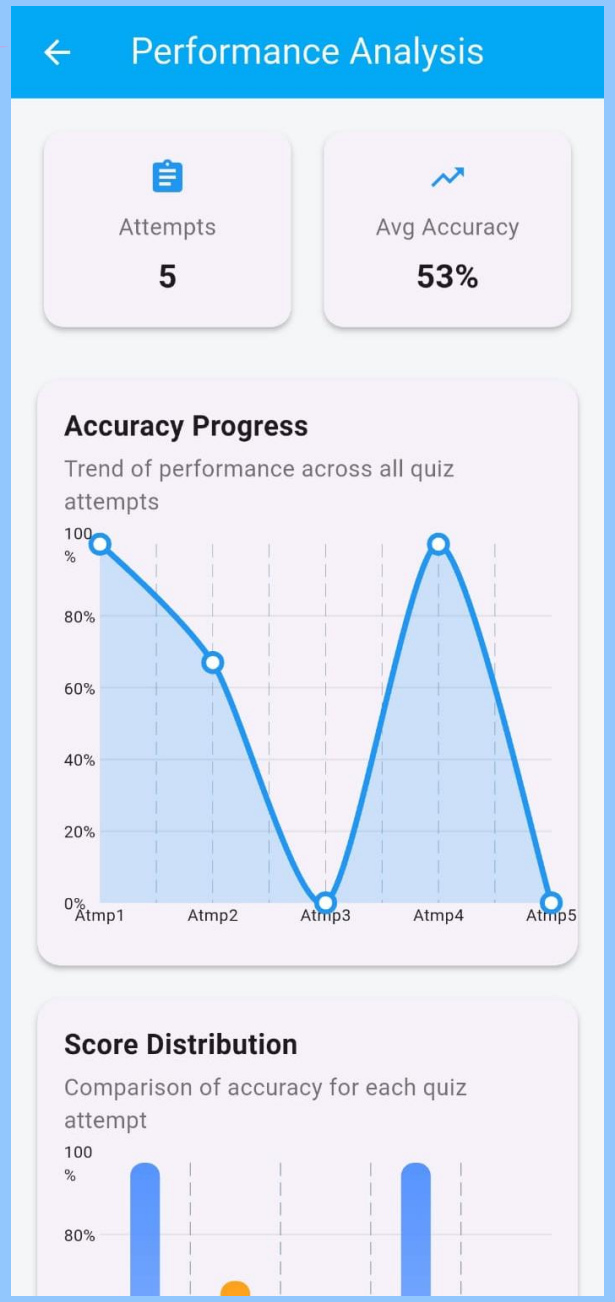
QUIZ FLOW


Quiz
2s
Question 1 of 3
Question
What is Algebra?
h
k
n
j

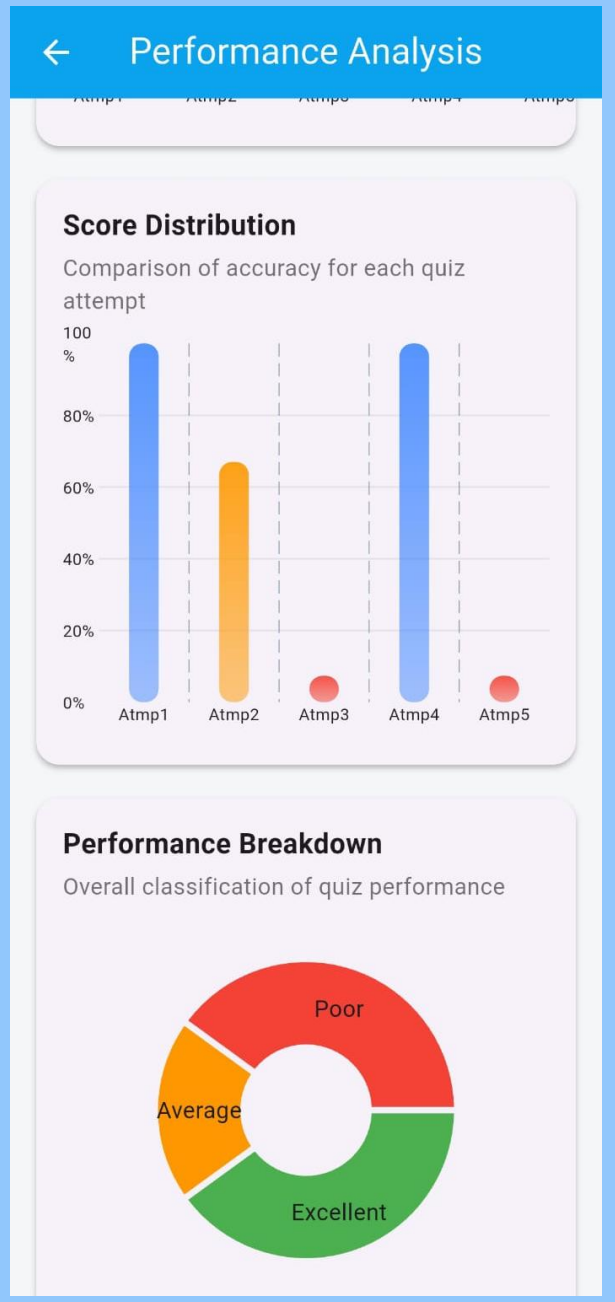



Quiz Result
Quiz Completed
You have successfully completed the quiz.
100%
Accuracy
Score
1 / 1
Status
Passed
View Performance Analysis
Back

PROGRESS & ANALYSIS

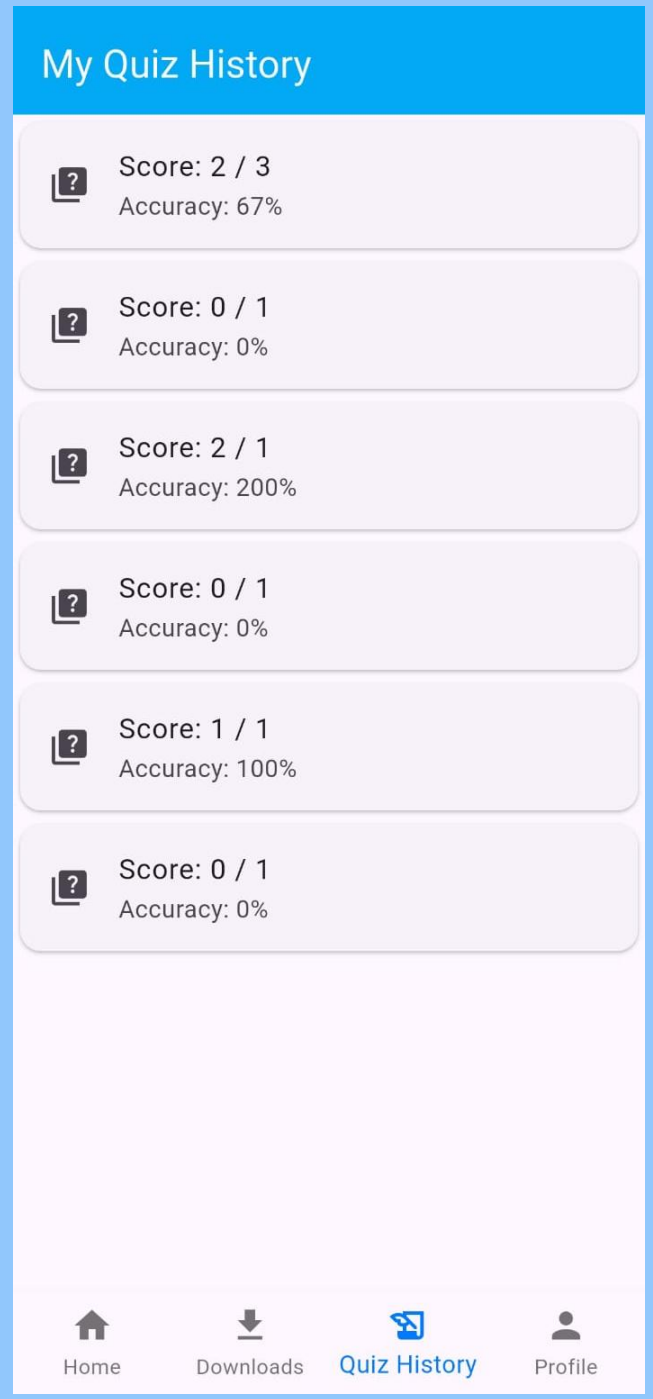


- Track student performance through quiz results.
- Provide visual insights (graphs/charts)
- Helps users identify strong and weak areas for improvement.



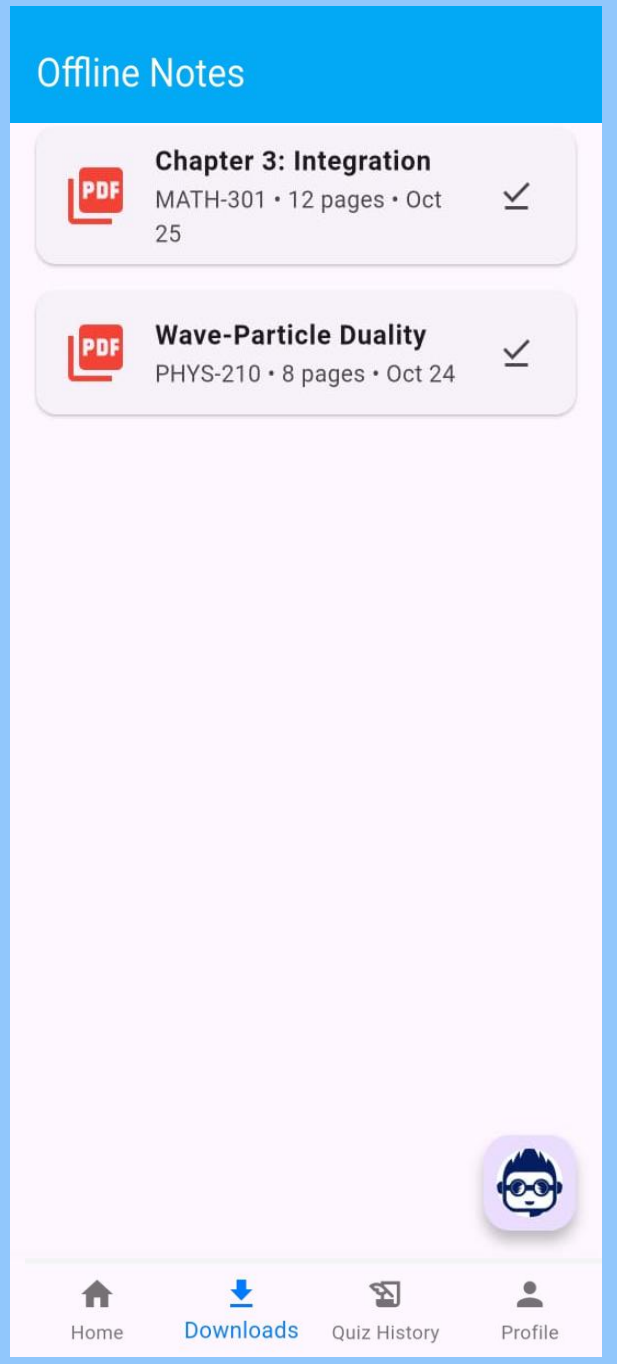
QUIZ HISTORY

- The system stores previous quiz attempts for future reference.
- Users can review past results to analyze their progress and identify weak areas.



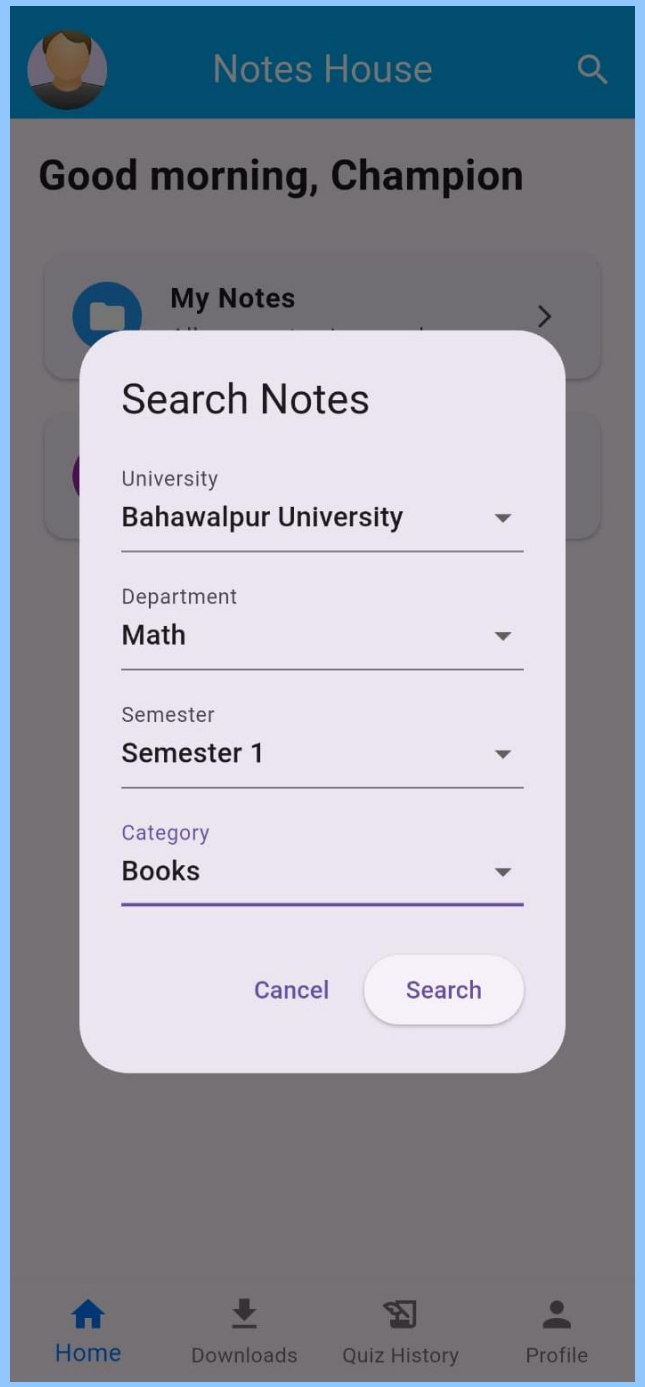
OFFLINE DATA ACCESS

- Downloaded content (Notes, Books, etc.) stays inside the app — like **YouTube Downloads**.
- Users can access their saved materials anytime without internet.



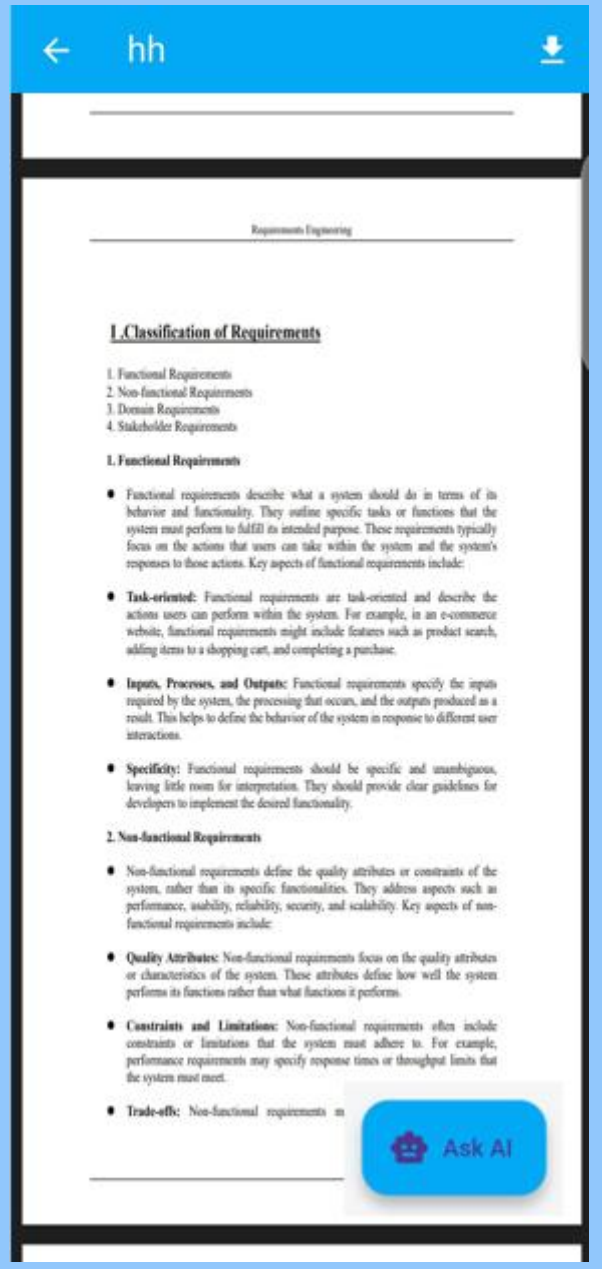
SMART SEARCH

- Enables users to **search directly** for notes or books.
- Eliminates the need to manually navigate through semesters and departments.
- Quick and user-friendly access to desired materials.



IN-APP DOCUMENT READING

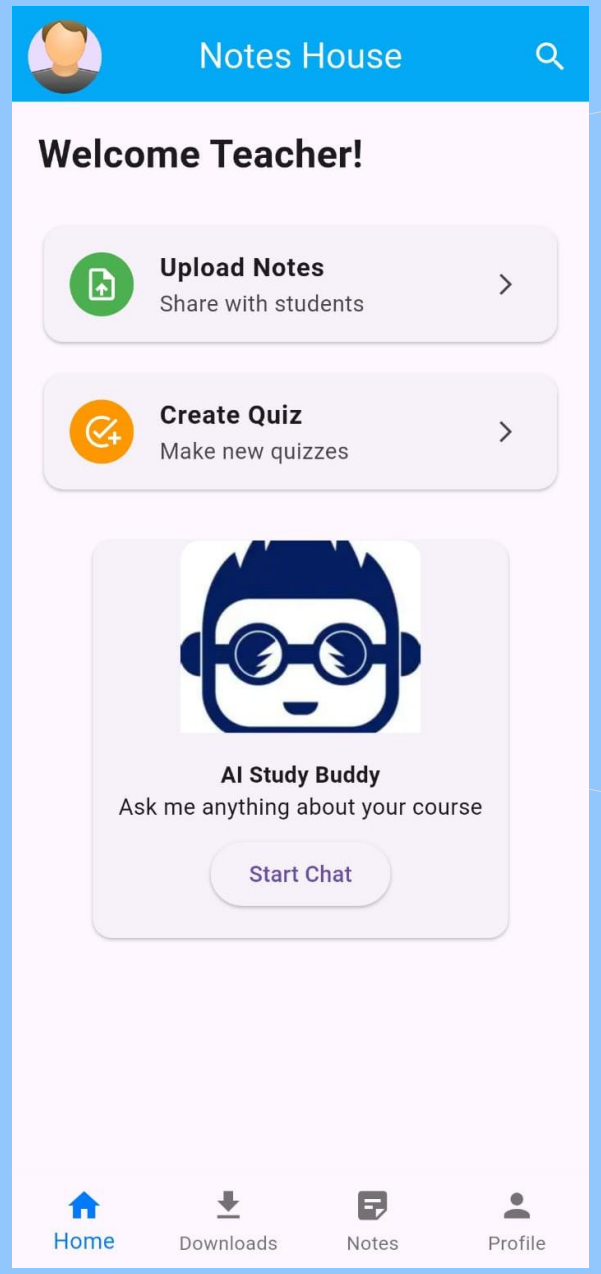
- Users can read documents directly in the app.
- Supports **highlighting** and **annotation** for better learning.
- Enhances focus and enables interactive learning.



Teacher DashBoard

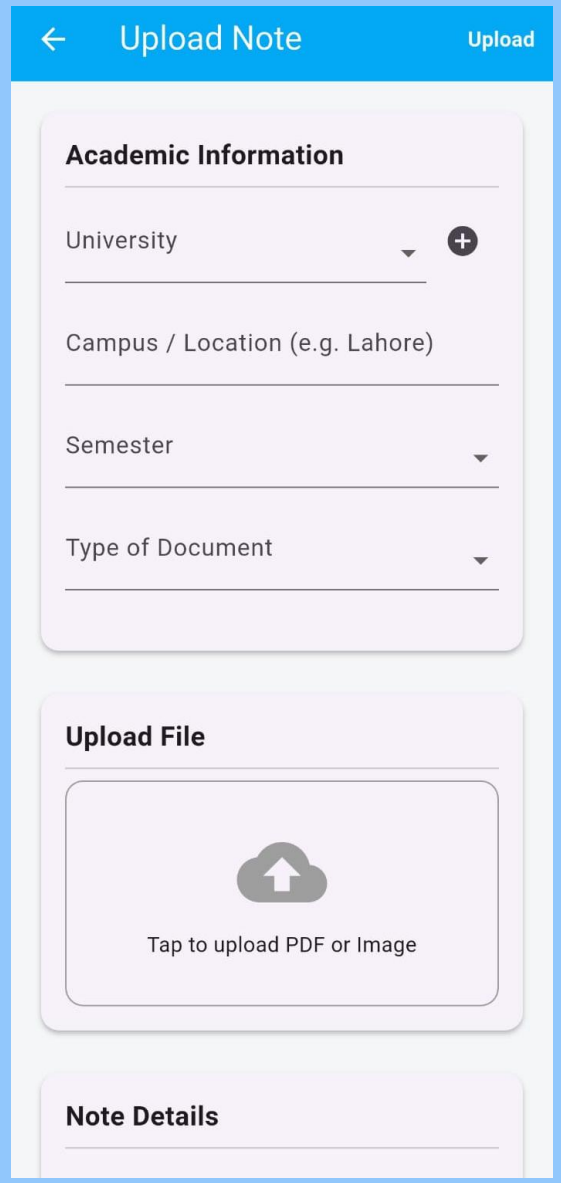
DASHBOARD (HOME-SCREEN)

- Upload notes and study material.
- Create quizzes.
- Access all shared resources.
- Simple and intuitive interface.

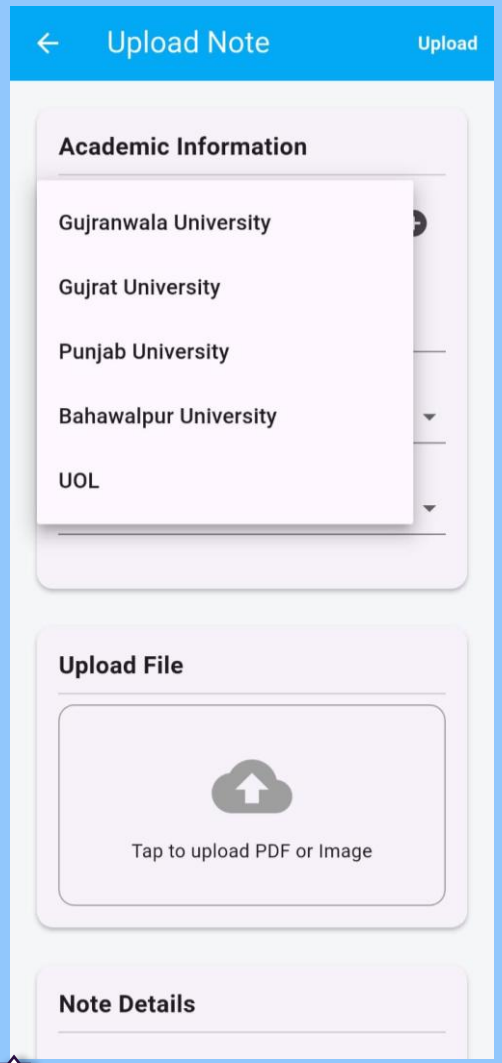


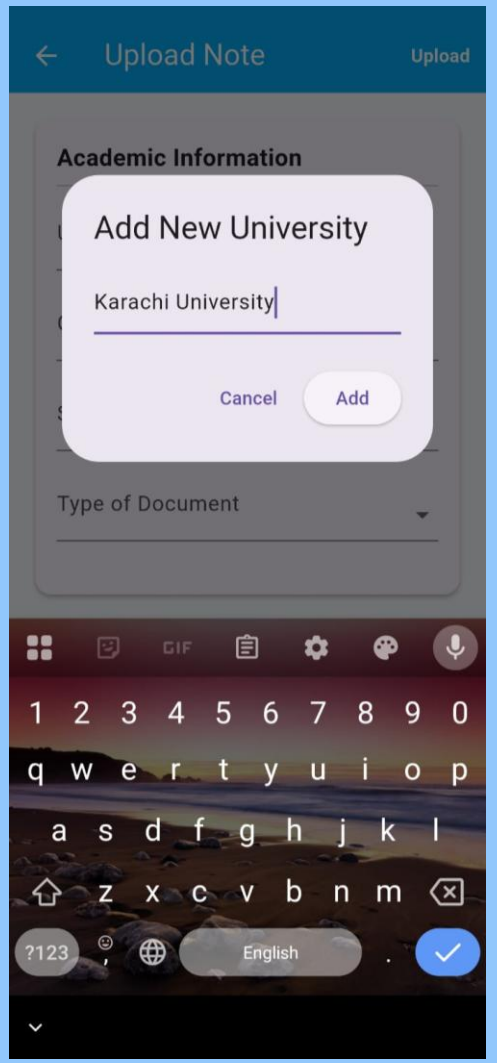
TEACHER-CONTRIBUTED NOTES

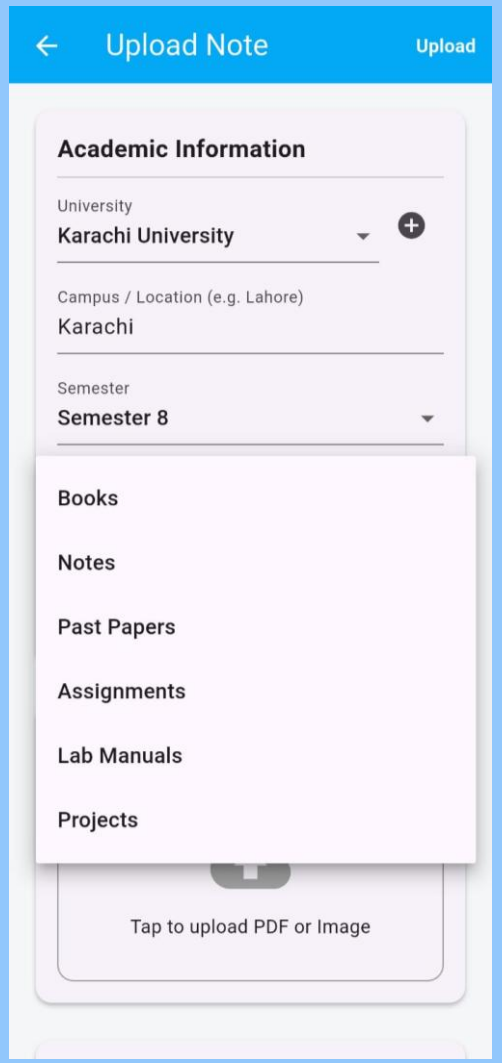
- Verified Teacher can upload notes.
- Teacher can choose University from list (already added).
- Teacher can Add University, if not exist in list.
- Teacher cad Add Department, if not in list.
- Admin approval ensures content quality before publishing.
- Notifications are sent to users when new content is added.



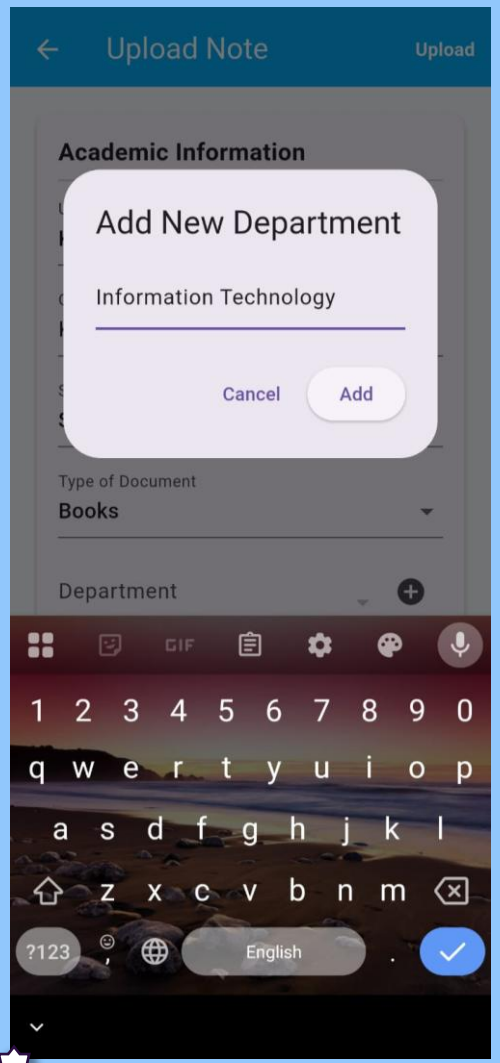
ADD NOTES FLOW

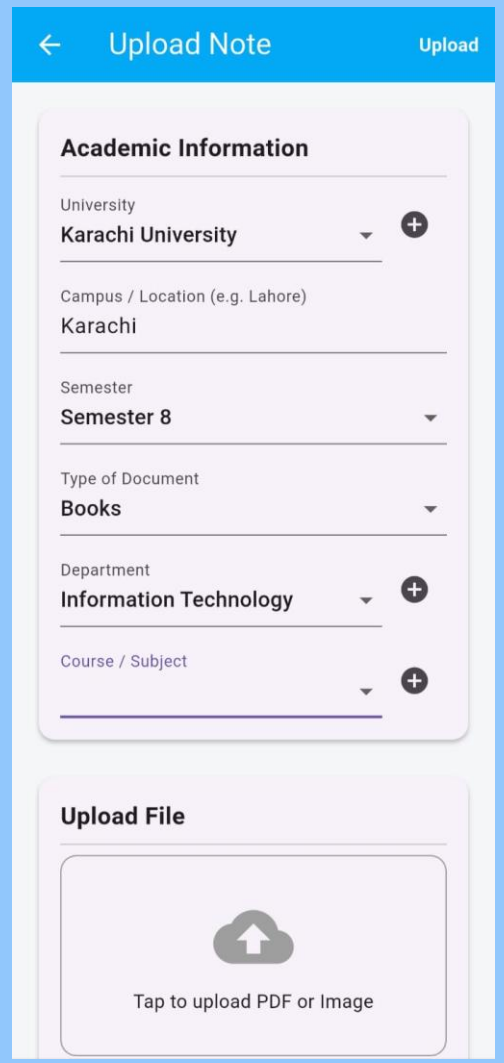


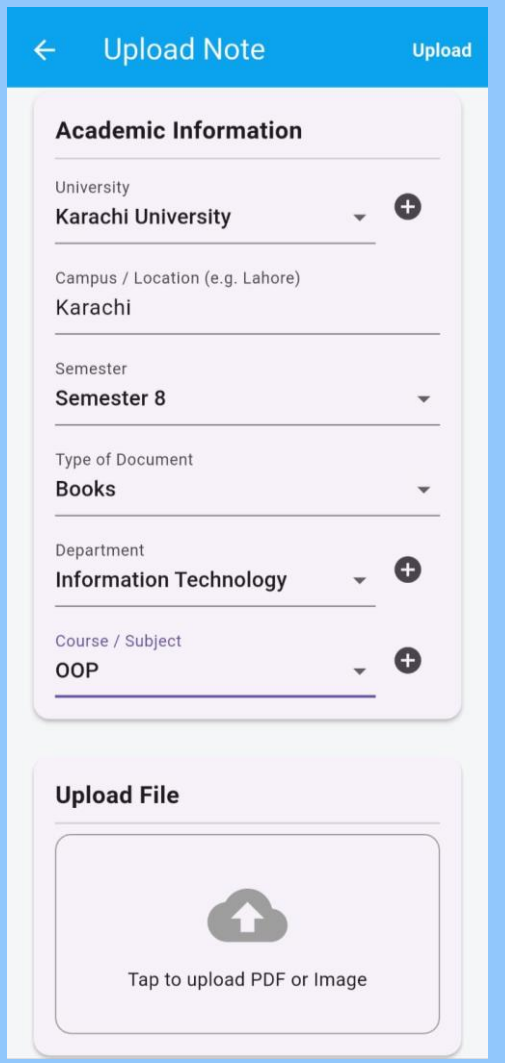




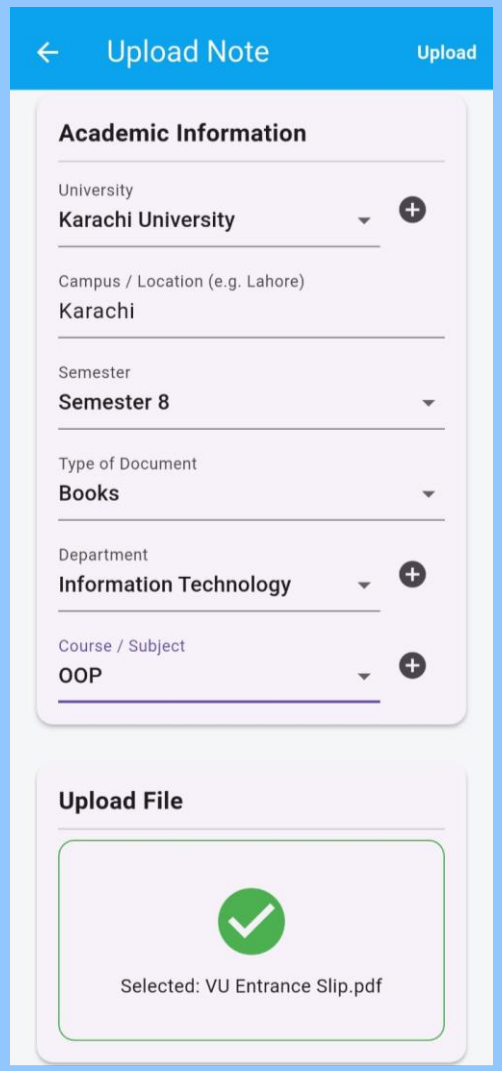
ADD NOTES FLOW

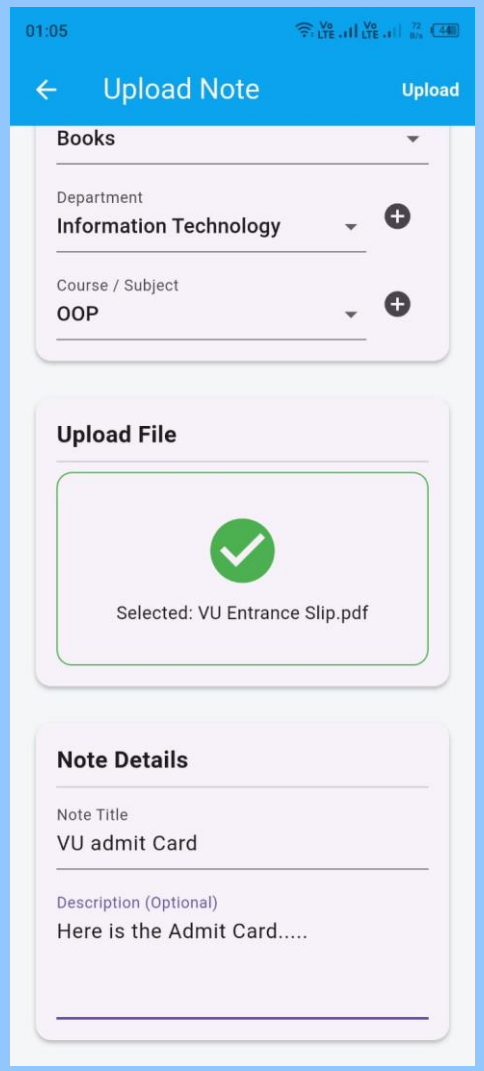


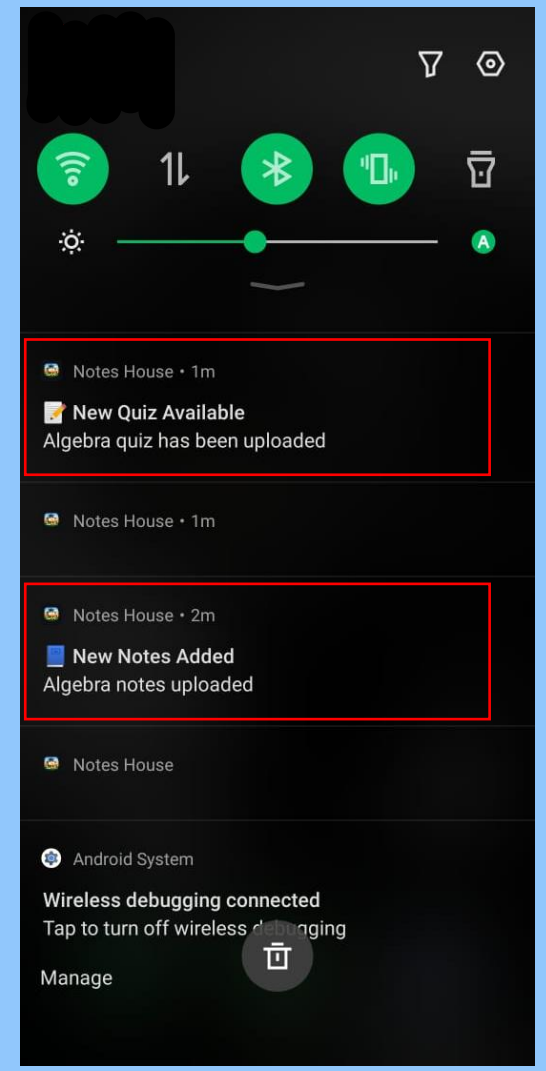




ADD NOTES FLOW

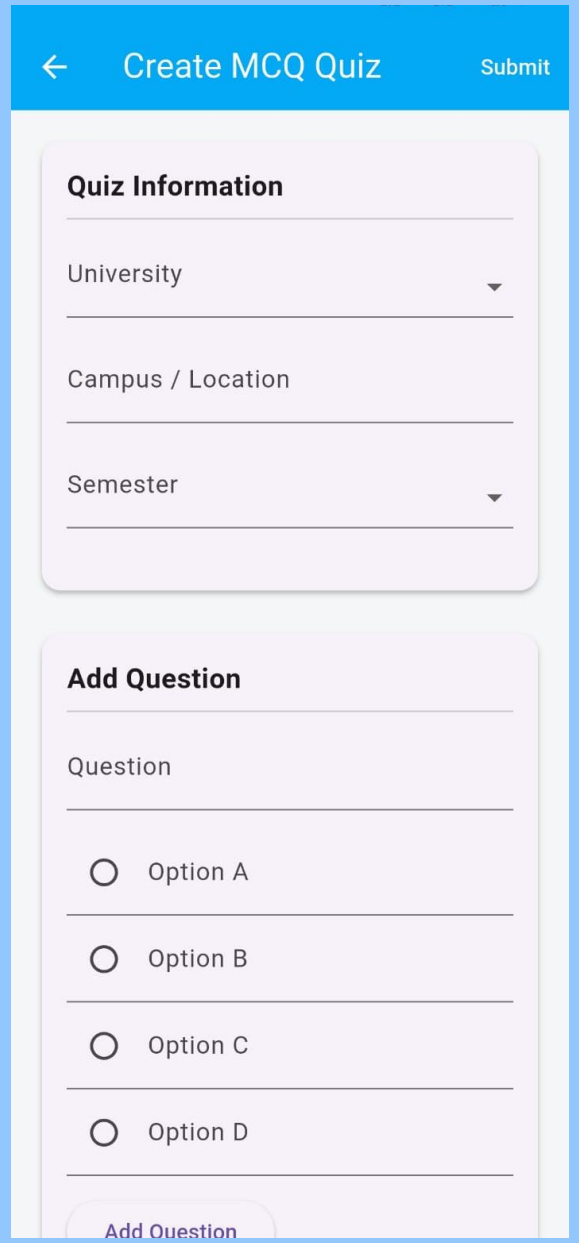




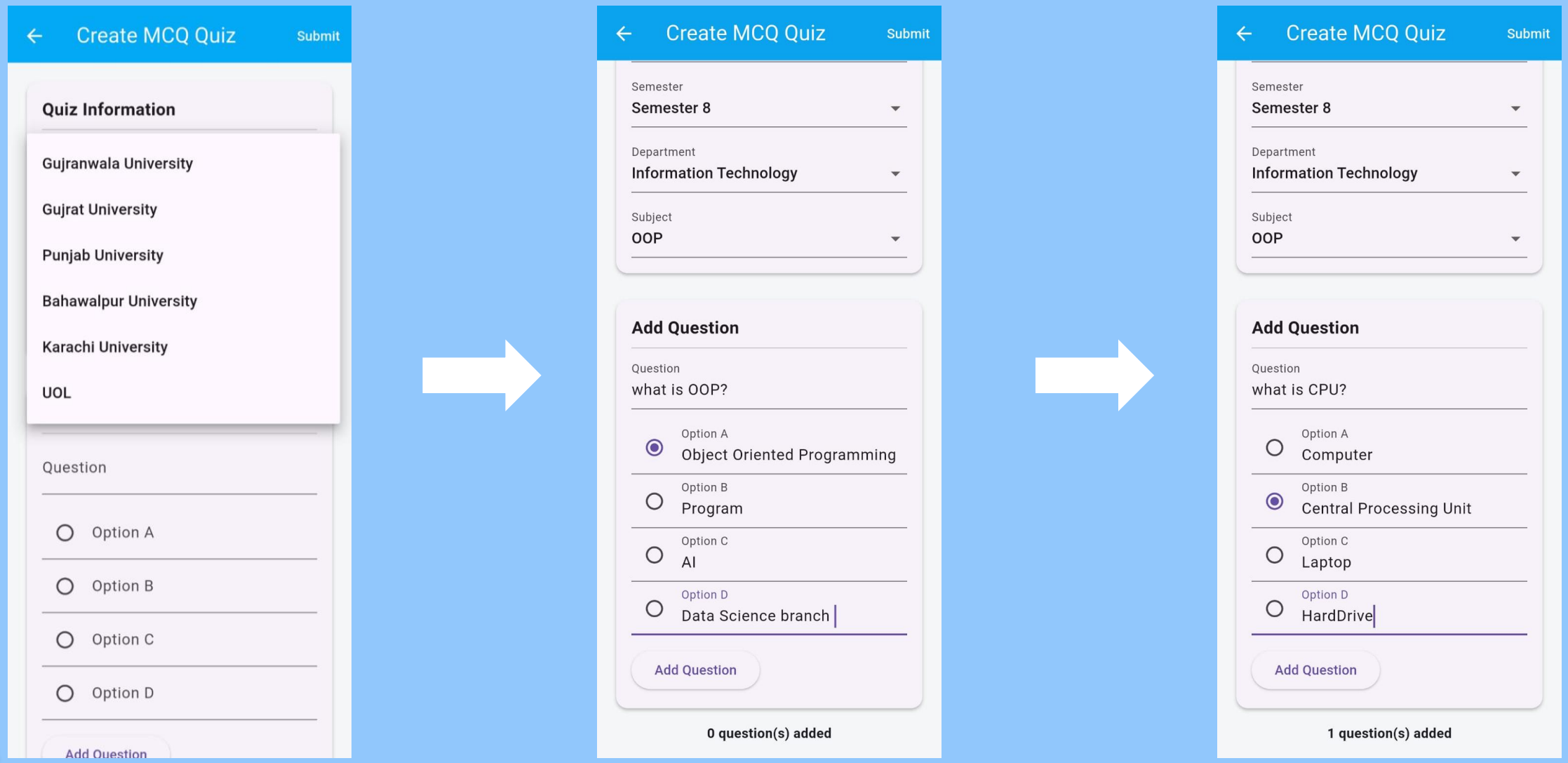


USER-CONTRIBUTED QUIZ

- Verified Teacher can upload quizzes.
- Teacher can choose University from list (already added).
- Admin approval ensures content quality before publishing.
- Notifications are sent to users when new content is added.



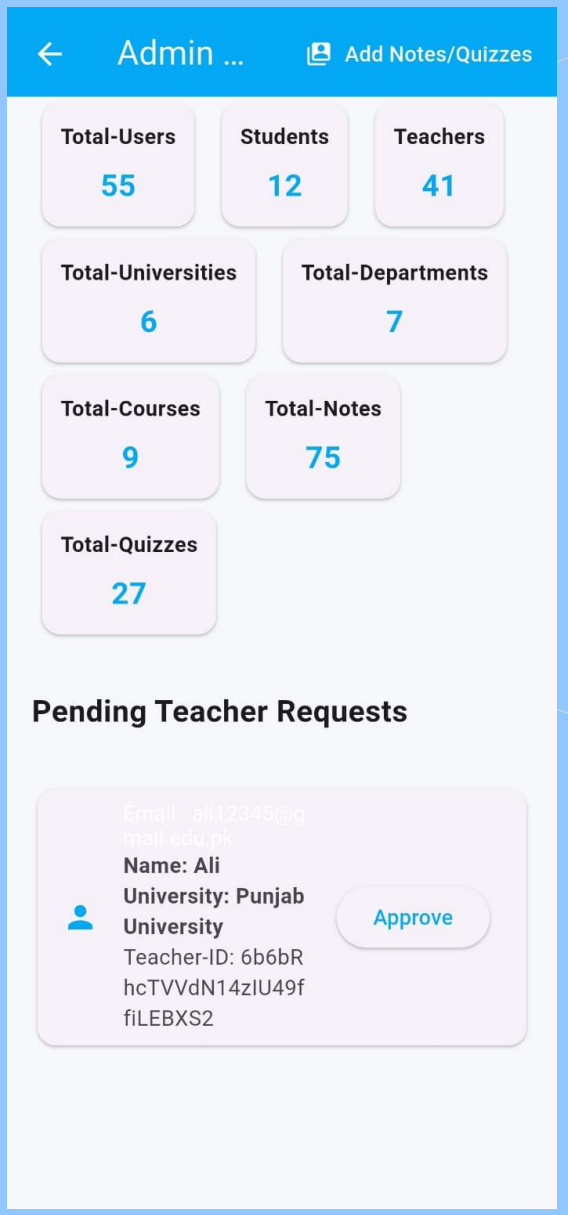
QUIZ CREATING FLOW



Admin DashBoard

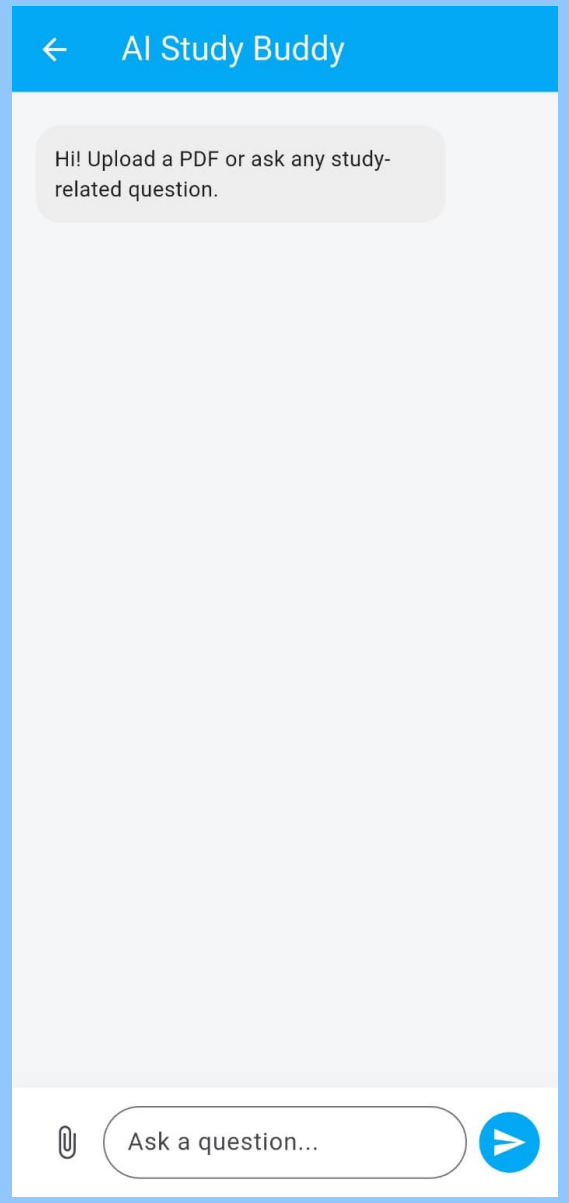
DASHBOARD (HOME-SCREEN)

- Admin manages teacher approvals and verifies uploaded content.
- Ensures system integrity and content quality.



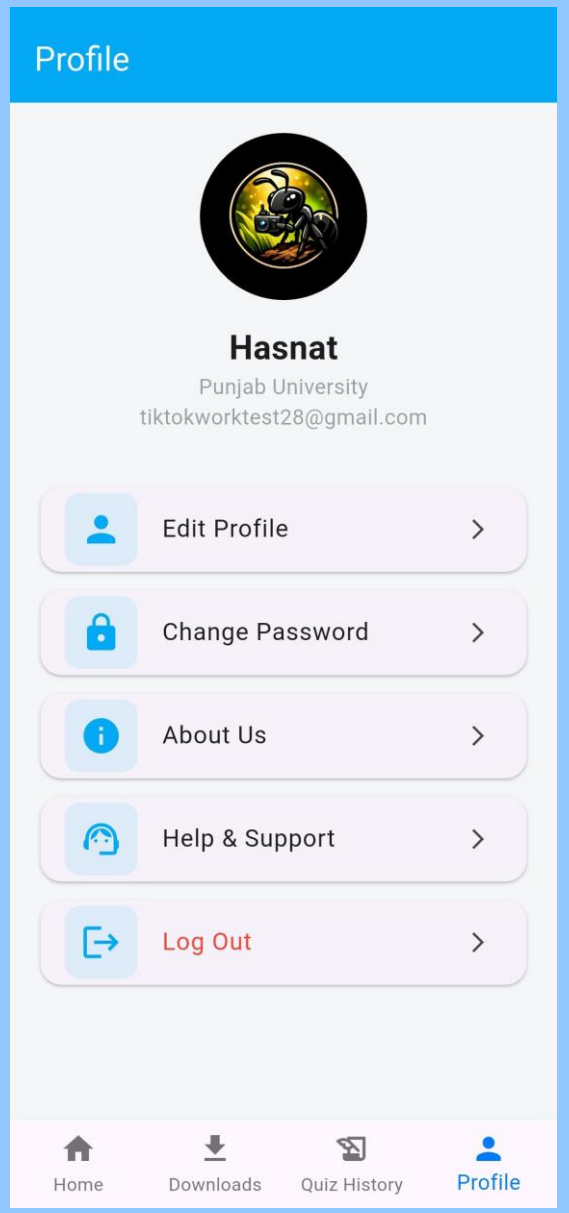
AI CHATBOT

- Provides document-based assistance by answering questions from uploaded notes.
- Ensures personalized and private learning experience for each user.



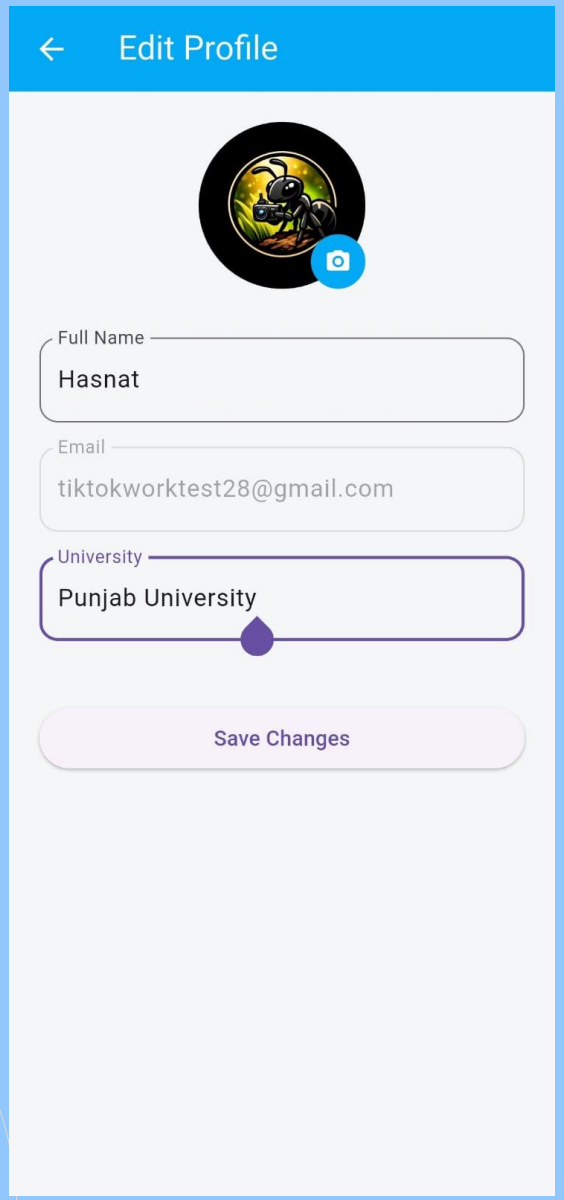
SETTINGS & PROFILE

- Allow users to manage account like:
  - Profile section for managing user account details
  - Change Password
  - About us
  - Help & Support
  - Login / Logout options



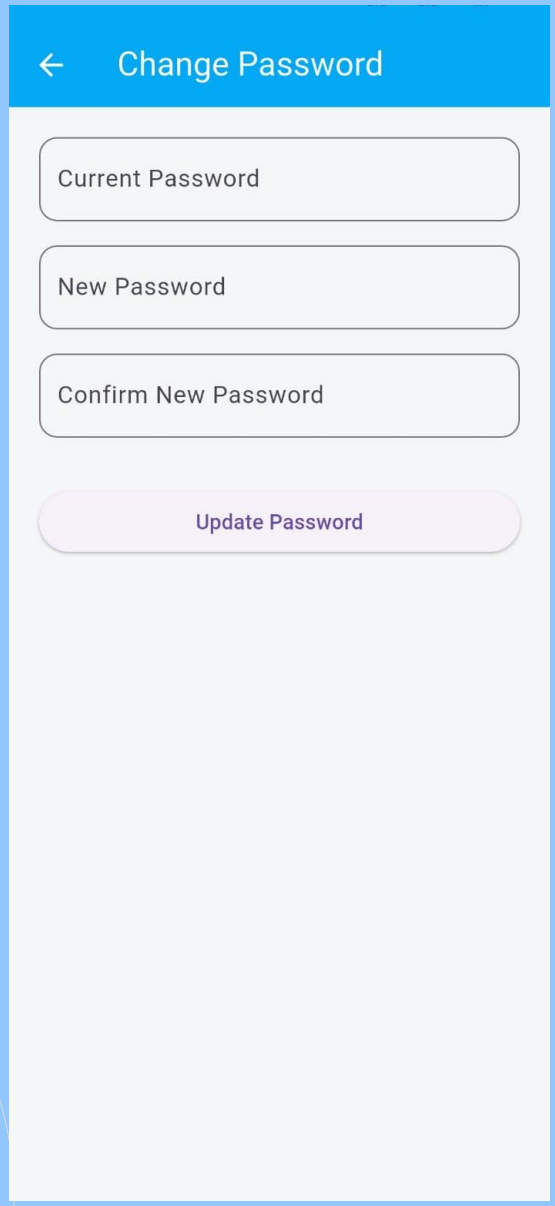
PROFILE EDIT

- Profile Edit include:
  - Name Change
  - University Detail



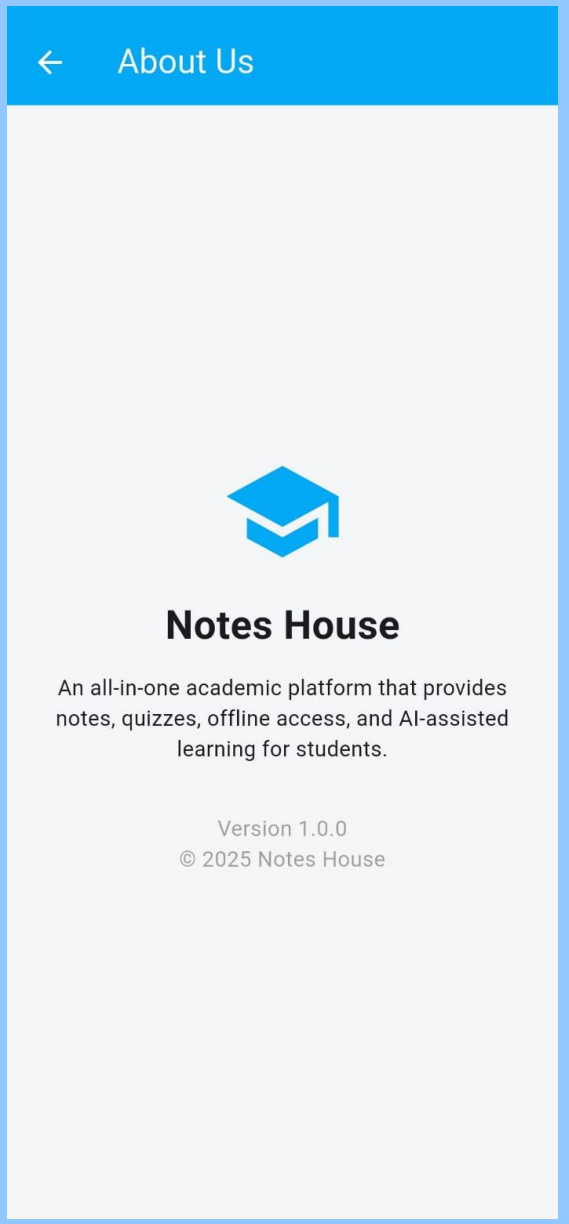
PASSWORD EDIT

- User can Change Password:
  - Enter Current Password
  - New Password
  - Confirm Password



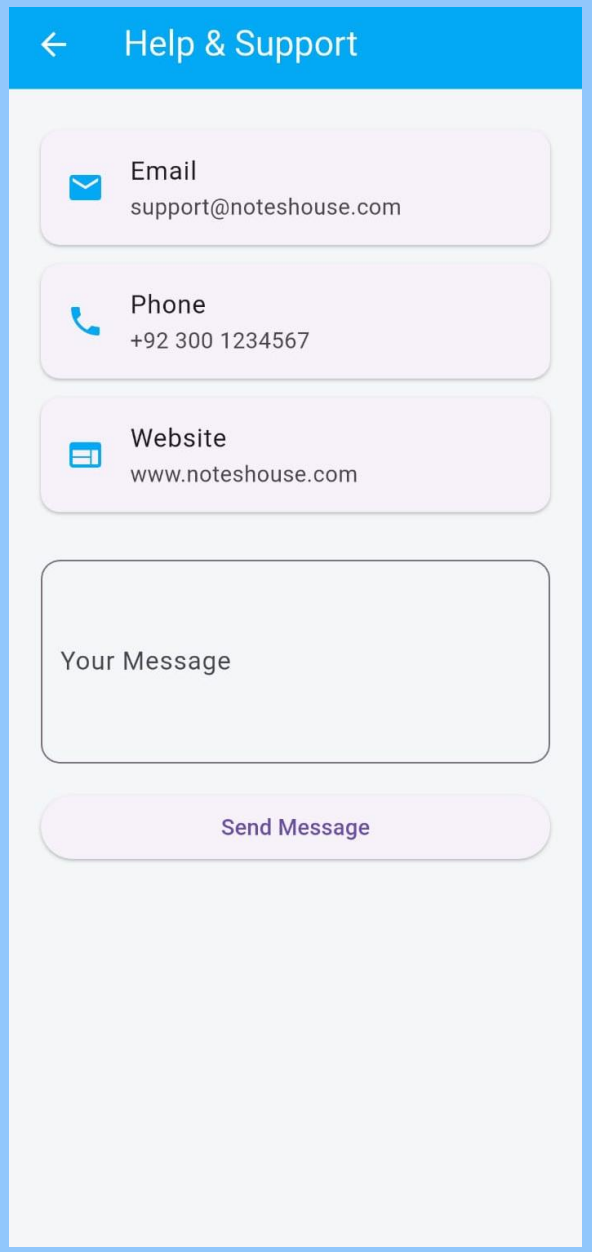
ABOUT US

- **FApp Crafters**
  - We are a team of passionate developers focused on improving the educational ecosystem in Pakistan.
  - Our goal is to provide a reliable, intelligent, and centralized platform that enhances student learning through technology and innovation.



HELP & SUPPORT

- User can Contact us:
  - Via Email
  - Phone (WhatsApp)
- User can provide feedback to improve the platform.




THANK YOU!